Model Collapse is a Singular Complexity Trajectory

Sarwesh Rauniyar
Independent Researcher
srauniy1@alumni.jh.edu

Abstract

Recursive training on synthetic data can drive generative models toward collapse, yet most explanations treat collapse as a statistical phenomenon: tail loss, variance shrinkage, or distributional drift. We develop a singular-geometric view of collapse. In a labeled Gaussian mixture abstraction of recursive self-training, we show that mode extinction induces a monotone trajectory in the real log-canonical threshold: as recursive resampling eliminates active modes, the overcomplete mixture moves through increasingly singular strata, with strictly decreasing singular complexity and a growing lower bound on Fisher-information nullity. We further show that this loss of identifiability begins before exact extinction: as a component's weight becomes small, Fisher curvature in its associated parameter directions has an upper bound proportional to its weight, under uniformly bounded component-score moments, so exact singularity is the endpoint of a continuous near-nullity trajectory. Thus collapse corresponds not only to reduced distributional diversity, but also to a progressive loss of locally identifiable parameter directions. This theory separates the singular geometry of collapse from the multinomial/Wright–Fisher absorption dynamics. We also prove that, in the mixture abstraction, injecting enough real data prevents component extinction and keeps the singular-complexity trajectory from descending over a fixed horizon with high probability. We then use the theory to motivate loss-landscape diagnostics for neural recursive training, including Hessian spectra, Gauss–Newton effective rank, decoder-block curvature, feature-rank concentration, and local learning coefficient estimates. Experiments span Gaussian mixtures, VaDE, standard VAEs, diffusion models up to 96M parameters, and OPT-125M. Matched fixed-real controls, early-warning prediction, and budget-matched interventions support recursion-specific geometric degeneration and practical prevention.

1 Introduction

Recursive training on synthetic data can cause generative models to lose rare modes, shrink tail mass, and drift from the original distribution [Shumailov et al., 2024, Bertrand et al., 2024, Dohmatob et al., 2024]. Existing explanations are mainly distributional, through variance shrinkage, tail loss, and divergence across generations [Shumailov et al., 2024, Alemohammad et al., 2024, Hataya et al., 2023], or probabilistic, through multinomial and Wright–Fisher absorption dynamics [Suresh et al., 2024, Borkar, 2025, Jangjoo et al., 2025, Benati et al., 2025, Riis, 2026]. We ask a complementary question: what happens to the *loss geometry* when recursive training eliminates modes?

Our thesis is that collapse is also a singularization process. When a mode disappears, its associated parameters become locally unidentifiable; before exact disappearance, small component weight already induces $O(\pi_k)$-scale Fisher curvature in component-specific directions. In a labeled Gaussian mixture abstraction, we show that mode extinction induces a monotone trajectory in the real log-canonical threshold (RLCT), the singular-learning-theory measure of effective complexity [Watanabe, 2009, 2018]. In VaDE-style neural latent-variable models [Jiang et al., 2017], we show that the

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

same mechanism appears as continuous latent-prior Fisher-curvature collapse, while shared decoder curvature is controlled by latent feature second moments. These results motivate loss-geometry diagnostics based on latent-prior Fisher curvature, local learning coefficient estimates [Furman and Lau, 2024], Hessian and Gauss–Newton spectra, decoder-block curvature, feature-rank concentration, and distributional mode coverage.

Contributions. We contribute: **(i)** a target-indexed RLCT trajectory theorem for labeled Gaussian-mixture recursion, plus a finite-horizon real-data protection condition $\alpha \geq \log(KG/\delta)/(n\pi^*_{\min})$; **(ii)** a neural latent-variable bridge showing that near-extinct VaDE components have $O(\pi_k)$-scale Fisher curvature, with exact Fisher nullity at extinction, and a decoder curvature identity linking final-layer Hessian rank to latent feature second moments; and **(iii)** empirical validation across Gaussian mixtures, VaDE, standard VAEs, and a compact DDPM, where synthetic-only recursion induces geometry degeneration, VaDE geometry diagnostics predict future component loss, and real-data injection stabilizes distributional and geometric collapse metrics. Further results cover matched fixed-real controls, full-model LLC, larger DDPMs and OPT-125M, budget-matched prevention, prospective dose transfer, Fisher/GGN bounds, and a conditional full-model RLCT criterion.

Related work. Prior work characterizes recursive model collapse through tail loss, variance shrinkage, divergence across generations, and probabilistic absorption in multinomial and Wright–Fisher-style processes. Separately, singular learning theory uses the RLCT and local learning coefficient to quantify effective complexity in singular models [Watanabe, 2009, 2018, Furman and Lau, 2024]. We connect these literatures through RLCT trajectories along recursive collapse and quantitatively small Fisher curvature in near-extinct latent components before exact singularity. This turns singular geometry into a pre-collapse diagnostic rather than only a post-extinction description. Full related work appears in the appendix.

2 Theory

2.1 Labeled Gaussian mixture recursion

We track component loss in a K-component unit-variance Gaussian mixture

$$p_\theta(x) = \sum_{k=1}^{K} \pi_k \varphi(x \mid \eta_k), \qquad \theta = (\pi, \eta_1, \ldots, \eta_K),$$

where $\varphi(\cdot \mid \eta)$ is the unit-variance Gaussian density with mean η. Starting from $\theta_0 = \theta^*$ with $\pi^*_k > 0$ for all k and distinct means, the labeled recursion draws

$$z_i^{(g)} \sim \mathrm{Categorical}(\pi_g), \qquad x_i^{(g)} \mid z_i^{(g)} = k \sim \varphi(\cdot \mid \eta_{g,k}).$$

Let $N_{g,k} = |\{i : z_i^{(g)} = k\}|$. The labeled MLE update is

$$\pi_{g+1,k} = \frac{N_{g,k}}{n}, \qquad \eta_{g+1,k} = \frac{1}{N_{g,k}} \sum_{i : z_i^{(g)} = k} x_i^{(g)} \quad (N_{g,k} \geq 1),$$

with $\eta_{g+1,k}$ irrelevant when $N_{g,k} = 0$. Let

$$E_g = \{k : \pi_{g,k} = 0\}, \qquad r_g = K - |E_g|.$$

The update $\pi_{g+1} = N_g/n$, with $N_g \sim \mathrm{Multinomial}(n, \pi_g)$, follows the neutral K-allele Wright–Fisher chain, linking support evolution to its martingale, monotone-extinction, and vertex-absorption properties. Oracle labels determine the update, while geometry uses the ordinary unlabeled mixture likelihood. The means remain distinct almost surely at each finite generation, which implies neither fixed separation nor one density mode per component.

2.2 The singular complexity trajectory

Component loss changes complexity through the model/target RLCT. For K nonnegative univariate Gaussian components with known unit variance and an r-component target [Kurumadani, 2026],

$$\lambda(K, r) = \frac{K + 3r - 2}{4}, \qquad 1 \leq r \leq K.$$

Assumption 1 (Model and target for the mixture RLCT)**.** *For each finite-generation target with r distinct positive-weight components, choose a compact interval I whose interior contains the target means. Evaluate the model-level coefficient on $\Delta^{K-1} \times I^K$, with a smooth prior density strictly positive up to the boundary relative to affine-weight and mean Lebesgue measure.*

Under Assumption 1, the proof in Appendix D gives

$$\lambda(K, K) = \frac{2K-1}{2}, \qquad \lambda(K, r) - \lambda(K, r-1) = \frac{3}{4} \quad (2 \le r \le K),$$

and we define the global, target-indexed trajectory RLCT by

$$\lambda_g := \lambda(K, r_g).$$

This minimum local RLCT over target representations in the overcomplete model need not equal the current fitted parameter's local coefficient (Appendix A.1). For curvature, write

$$I(\theta) := \mathbb{E}_{x \sim p_\theta} \left[\nabla_\theta \log p_\theta(x) \nabla_\theta \log p_\theta(x)^\top \right]$$

for the Fisher information in $K-1$ affine weight coordinates and the component means.

Theorem 1 (Singular complexity trajectory in labeled Gaussian mixture recursion)**.** *Under Assumption 1, in the labeled synthetic-only recursion, λ_g is non-increasing almost surely. Losing $d \ge 1$ components in one update decreases λ_g by $3d/4$. The process is eventually absorbed at a single surviving component, so $\lambda_g \to \lambda(K, 1)$. Moreover,*

$$\dim \ker I(\theta_g) \ge |E_g|, \qquad \liminf_{g \to \infty} \dim \ker I(\theta_g) \ge K - 1.$$

Theorem 1 connects classical absorption dynamics to singular geometry. Extinction persists in the Wright–Fisher chain, while the RLCT formula assigns each component loss a $3/4$ drop toward the single-component endpoint $\lambda(K, 1)$. Each extinct component's mean score also vanishes, giving a Fisher-null direction (Appendix B). Prior probabilistic analyses track mixture weights through extinction and absorption (Appendix A). We track each extinction's geometric effect through its exact RLCT decrement and Fisher-null mean directions.

Sufficient real-data injection preserves support and prevents this descent over a finite horizon with high probability.

Proposition 1 (Finite-horizon prevention under real-data injection)**.** *In the labeled α-mixed recursion, each generation uses $\lfloor (1-\alpha)n \rfloor$ labeled synthetic samples from p_{θ_g} and $\lceil \alpha n \rceil$ labeled real samples from $p^* = p_{\theta^*}$. Let $N_{g,k}^{(\alpha)}$ be the total number of mixed training samples assigned to component k, and assume $\theta_0 = \theta^*$, $\alpha \in [0, 1]$, and $\pi_k^* \ge \pi_{\min}^* > 0$. For an integer horizon $G \ge 1$ and $\delta \in (0, 1)$,*

$$\mathbb{P}\Big(\exists g < G, \exists k : N_{g,k}^{(\alpha)} = 0 \Big) \le KG \exp\{-\alpha n \pi_{\min}^*\}.$$

Consequently, if

$$\alpha \ge \frac{\log(KG/\delta)}{n \pi_{\min}^*},$$

then with probability at least $1 - \delta$, no component becomes extinct over the horizon and

$$\lambda_g = \lambda(K, K) = \frac{2K-1}{2}, \qquad 0 \le g \le G.$$

Appendix E.2 proves this and the sharper $m_\alpha = \lceil \alpha n \rceil$ bound. If the sufficient fraction exceeds one, the guarantee is infeasible at this sample size and cannot be restored by clipping the fraction to one.

2.3 From mixtures to neural generative models

We next connect neural component loss and feature concentration to curvature, then give a conditional RLCT criterion for complexity decrease. Appendix H gives the statements and proofs.

Component curvature and full-Fisher near-nullity. In VaDE, component weights bound component-specific Fisher curvature. Write $p_\Theta(x) = \sum_k \pi_k q_k(x)$, where $q_k(x) = \int p_\psi(x \mid z)\mathcal{N}(z \mid \mu_k, \Sigma_k)\, dz$, and let $\rho_k = \pi_k q_k / p_\Theta$. The mean/covariance block a_k has score $\rho_k s_k$, with $s_k = \nabla_{a_k} \log q_k$, so the condition $\mathbb{E}_{q_k}[s_k s_k^\top] \preceq C_k I$ gives $I_{a_k a_k} \preceq \pi_k C_k I$. For a nonempty set E of small-weight components, Proposition 4 gives the full-Fisher bound

$$\lambda^{\uparrow}_{d_E}(I(\Theta)) \le \delta_E, \qquad d_E = \sum_{k \in E} \dim(a_k), \qquad \delta_E = \max_{k \in E} \pi_k C_k.$$

Thus at least d_E full-Fisher eigenvalues are at most δ_E, becoming null at exact extinction, with $d + d(d+1)/2$ directions per full-covariance component in latent dimension d. The bound vanishes when $\delta_E \to 0$; since δ_E also depends on C_k, small weights alone need not suffice. A uniform positive lower bound on latent covariance suffices in direct Gaussian mean/covariance coordinates. The $O(\pi_k)$ bound controls how large curvature can remain, without imposing a proportionality law.

Softmax and unlabeled recursion. For softmax weights, finite logits describe the simplex interior, with extinction reached only as a boundary limit. Holding component marginals fixed gives

$$\nabla_\beta \log p_\Theta = \rho - \pi, \qquad I_\beta = \mathrm{Cov}(\rho(X)) \preceq \mathrm{Diag}(\pi) - \pi\pi^\top.$$

The bound $I_{\beta_k \beta_k} \le \pi_k(1 - \pi_k)$ links small weights to small logit curvature. Rank comparisons exclude the always-null constant-shift logit gauge. Locally invertible coordinate changes preserve Fisher rank, although curvature magnitudes and finite-sample LLC estimates need not be preserved.

These scores also connect curvature to unlabeled recursion. For the ideal update $\pi_{g+1} = n^{-1} \sum_i \rho_g(X_i)$ with conditionally independent $X_i \sim p_{\Theta_g}$ given $\mathcal{F}_g$, define $D(\pi) = 1 - \|\pi\|_2^2$. Then

$$\mathbb{E}[D(\pi_{g+1}) \mid \mathcal{F}_g] = D(\pi_g) - n^{-1} \operatorname{tr} I_{\beta,g}.$$

The Fisher trace determines the expected one-step loss of weight diversity. The weights are martingales with $\sum_g \operatorname{tr} I_{\beta,g} < \infty$ almost surely, and uniform separation yields $D(\pi_g) \to 0$. This responsibility/Fisher identity needs no oracle labels, but trained VaDE adds a one-step optimization-error term. Oracle labels are required for exact multinomial updates.

Representation and conditional complexity. Representation geometry offers a second route to low curvature. For an output Jacobian J_W and loss curvature $0 \preceq B \preceq LI$, Proposition 5 bounds the generalized Gauss–Newton block by

$$G_W = \mathbb{E}[J_W^\top B J_W] \preceq L\mathbb{E}[J_W^\top J_W].$$

Small Jacobian-Gram eigenvalues thus imply small GGN eigenvalues, even though shared decoder parameters need not become Fisher-null when a latent component disappears. For a final decoder layer $f_W(z) = W\phi(z)$, the reconstruction loss and Hessian make this connection explicit (Appendix E.3):

$$\mathcal{L}_g(W) = \frac{1}{2\sigma^2}\mathbb{E}_{(z,x)\sim Q_g}\|W\phi(z) - x\|_2^2, \qquad \nabla^2_{\mathrm{vec}(W)}\mathcal{L}_g(W) = \frac{1}{\sigma^2}\mathbb{E}_{Q_g}[\phi(z)\phi(z)^\top] \otimes I_{\mathrm{out}},$$

where Q_g is the generation-g latent–data distribution, I_{out} is the output-dimensional identity, and Q_g, ϕ, and σ^2 are fixed while differentiating. Recursive feature concentration can therefore degenerate decoder curvature through the uncentered second moment, which differs from centered feature covariance by the feature-mean outer product. Convolutional blocks require the full-output Jacobian, including downstream derivatives. Effective-rank decline captures spectral concentration, so it alone does not establish vanishing absolute eigenvalues.

For complexity, on a common analytic neighborhood with comparable positive priors, $K_{g+1} \le CK_g$ implies $\lambda_{g+1} \le \lambda_g$. Deleting d transverse regular residual directions in a local product decomposition gives an exact $d/2$ drop, while the empirical LLC trajectories below test the broader interpretation.

We assess **P1**, GMM/VaDE Fisher collapse, and **P2**, exact GMM RLCT descent or empirical LLC decay. Across all four tiers, **P3** tracks Hessian/Gauss–Newton and feature-rank concentration, while **P4** tests stabilization under real-data injection. The held-out early-warning test **P5** then asks whether geometry predicts subsequent coverage loss. For standard VAEs, P2–P4 are empirical signatures. The compact DDPM tests P3–P4, while larger DDPMs and OPT also measure LLC.

Table 1: **Endpoints under recursion, fixed-real training, and injection.** LLC is full-model and DDPM and OPT ranks are normalized, with variability of one seed standard deviation. Lower NLL, perplexity, and FID are better. Appendix I gives full diagnostic endpoints and confidence intervals.

Model / metric	Initial	Synthetic	Fixed-real	$\alpha = 0.25$
VaDE / components	10.0 ± 0.0	3.6 ± 0.8	9.8 ± 0.3	8.9 ± 0.6
VaDE / feature rank	8.9 ± 0.4	3.9 ± 0.7	8.6 ± 0.5	—
VaDE / LLC	128.4 ± 6.7	80.1 ± 8.9	124.9 ± 5.9	119.6 ± 6.2
VaDE / held-out NLL	86.2 ± 1.4	121.8 ± 4.9	87.1 ± 1.6	—
VAE / entropy	2.30 ± 0.0062	1.21 ± 0.18	2.26 ± 0.04	> 2.05
VAE / feature rank	8.6 ± 0.5	3.7 ± 0.6	8.1 ± 0.4	7.6 ± 0.5
VAE / LLC	116.2 ± 5.8	69.4 ± 7.6	112.8 ± 6.1	108.9 ± 5.7
VAE / held-out NLL	101.4 ± 2.2	149.6 ± 6.1	103.0 ± 2.5	—
DDPM-1.8M / entropy	2.29 ± 0.02	1.37 ± 0.05	2.24 ± 0.04	2.28 ± 0.05
DDPM-1.8M / FID	33.2 ± 1.1	90.0 ± 2.3	34.1 ± 1.5	29.9 ± 1.0
DDPM-1.8M / feature rank	0.94 ± 0.02	0.51 ± 0.02	0.93 ± 0.02	0.96 ± 0.02
DDPM-1.8M / GGN rank	0.92 ± 0.03	0.53 ± 0.04	0.88 ± 0.03	0.89 ± 0.03
DDPM-12M / LLC	214.7 ± 9.8	127.5 ± 11.6	207.9 ± 10.2	198.6 ± 9.4
OPT-125M / LLC	286.3 ± 11.4	174.8 ± 13.7	276.5 ± 10.9	272.1 ± 10.8
OPT-125M / perplexity	21.4 ± 0.6	34.7 ± 1.8	21.9 ± 0.7	22.3 ± 0.8
OPT-125M / rare tokens	0.76 ± 0.02	0.41 ± 0.04	0.73 ± 0.02	0.72 ± 0.02
OPT-125M / hidden rank	0.91 ± 0.01	0.58 ± 0.03	—	0.88 ± 0.02
OPT-125M / GGN rank	0.89 ± 0.02	0.61 ± 0.04	—	0.86 ± 0.03

3 Experiments

We test (P1)–(P4) from exact theory to increasingly general models. Univariate Gaussian mixtures instantiate Theorem 1, while VaDE tests latent-prior Fisher scaling and decoder curvature. Standard VAEs test latent-variable geometry without explicit mixtures, followed by a compact DDPM beyond VAE-style models. Scale and transfer tests use 12M, 40M, and 96M DDPMs on CIFAR-10 and OPT-125M on WikiText-2. Real-data injection is a counterfactual control linking the finite-horizon guarantee for labeled mixtures to stabilization in neural experiments.

3.1 Setup and diagnostics

We separate recursion from training alone using three protocols. Recursive training follows

$$\mathcal{D}_{g+1} \sim p_{\theta_g}^{\otimes n}, \qquad \theta_{g+1} = \mathrm{Train}(\mathcal{D}_{g+1}, \theta_g),$$

using labeled MLE for Gaussian mixtures, ELBO maximization for VaDE and VAE, and denoising-score training on generated images for DDPMs. The fixed-real control reuses one real dataset, matching initialization, architecture, optimizer, learning rate, minibatch size, updates, evaluation split, checkpoints, and seeds, isolating continued optimization without new observations. The α-mixed control replaces an α-fraction of each generation's samples with fresh original-distribution samples, reaching fresh-real sampling at $\alpha = 1$.

Distributional diagnostics are active component count (mixtures and VaDE), class-coverage entropy (images), held-out negative log-likelihood where available, and FID (diffusion). We track geometry through empirical LLC, Hessian and Gauss–Newton spectral concentration, componentwise latent-prior Fisher curvature and near-nullity, decoder feature-covariance rank, and U-Net feature and blockwise Gauss–Newton ranks. VaDE and VAE use $n = 5000$, $G = 50$, and 10 seeds, while DDPMs use $G = 20$ and 5 seeds. Values are means $\pm$ one seed standard deviation unless otherwise stated. Protocols and additional results are in Appendices G and I.

Diagnostic glossary. Coverage entropy is Shannon entropy of generated class frequencies from a classifier trained only on real data. Active classes exceed 1% prevalence, and active VaDE components have weight $\geq 10^{-3}$. Fisher near-nullity tracks small curvature eigenvalues, whereas entropy effective rank $\exp(-\sum_i \widetilde{\lambda}_i \log \widetilde{\lambda}_i)$ measures spectral concentration. Feature and generalized Gauss–Newton

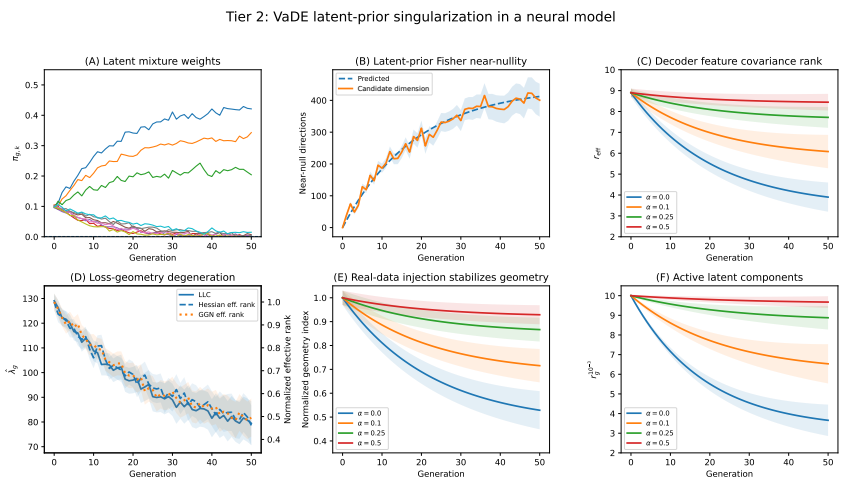


Figure 1: **VaDE latent-prior singularization under recursion.** On MNIST ($K = d = 10$, $n = 5000$, $G = 50$), synthetic-only recursion (A) drives latent weights near extinction, with (B) candidate low-curvature subspace dimension $|E_g^\varepsilon|(d + d(d+1)/2)$. Decoder feature-covariance effective rank (C) falls with latent support, while (D) empirical LLC and Hessian and Gauss–Newton effective ranks also decline. (E) Injection stabilizes aggregate geometry, while (F) active counts show largely preserved support at $\alpha \geq 0.25$. Finite-softmax weights approach the boundary continuously, unlike exact GMM extinction, allowing curvature, LLC, and effective ranks to change before threshold crossing. The 10-seed means smooth integer counts and are shown with 1σ bands.

(GGN) ranks probe representations and loss geometry, respectively. LLC gives distinct localized, finite-sample SGLD estimates of block-local and full-model complexity.

3.2 Tier 1 — Gaussian mixtures: extinction and singular geometry

Labeled GMMs connect observed component loss directly to singular geometry. With $K \in \{3, 5, 8\}$, $n \in \{50, 100, 200, 500, 1000\}$, and 10 seeds, each absorbing loss increases the Fisher-nullity lower bound and lowers the RLCT by Theorem 1. Fixed-data labeled MLE is stationary after its first fit, separating these steps from continued fitting. Injection at $\alpha \in \{0.25, 0.5\}$ preserves all five components through $G = 200$ for $K = 5, n = 100$ in the reported runs (Appendix F).

3.3 Tier 2 — VaDE: latent-prior singularization in a neural model

VaDE tests component loss and curvature in a learned mixture prior on MNIST, using $K = 10$ latent components, latent dimension $d = 10$, and $\alpha \in \{0, 0.1, 0.25, 0.5\}$.

Components with $\pi_{g,k} < 10^{-3}$ are treated as near-extinct:

$$E_g^\varepsilon = \{k : \pi_{g,k} < 10^{-3}\}, \qquad r_g^\varepsilon = K - |E_g^\varepsilon|, \qquad \kappa_{g,k} = \frac{\operatorname{tr} I_{a_k a_k}(\Theta_g)}{d + d(d+1)/2},$$

where $a_k = (\mu_{g,k}, \Sigma_{g,k})$ and $\kappa_{g,k}$ is normalized componentwise latent-prior Fisher curvature. Recursion reduces active component count even without oracle labels, while fixed-real training and injection largely preserve it (Figure 1, Table 1).

Across generations, components, and seeds, log–log regression links lower component weight to weaker curvature, consistent with the $O(\pi_{g,k})$ upper bound in Section 2.3 (Figure 5B). Each selected full-covariance component contributes $d + d(d+1)/2 = 65$ directions to the threshold-selected candidate low-curvature subspace of dimension $65|E_g^\varepsilon|$ (Table 5). Feature rank and LLC also fall (Table 1), while Hessian and Gauss–Newton effective ranks roughly halve. Injection at $\alpha \geq 0.25$ keeps support, feature rank, and aggregate geometry closer to initialization, linking support preservation to geometric stabilization (P1)–(P4).

Softmax coordinates provide a further curvature check. Logit-Fisher diagonals stay below $\pi_k(1 - \pi_k)$ at every checkpoint, and gauge-adjusted checks preserve the rank trajectories (Appendix I.4).

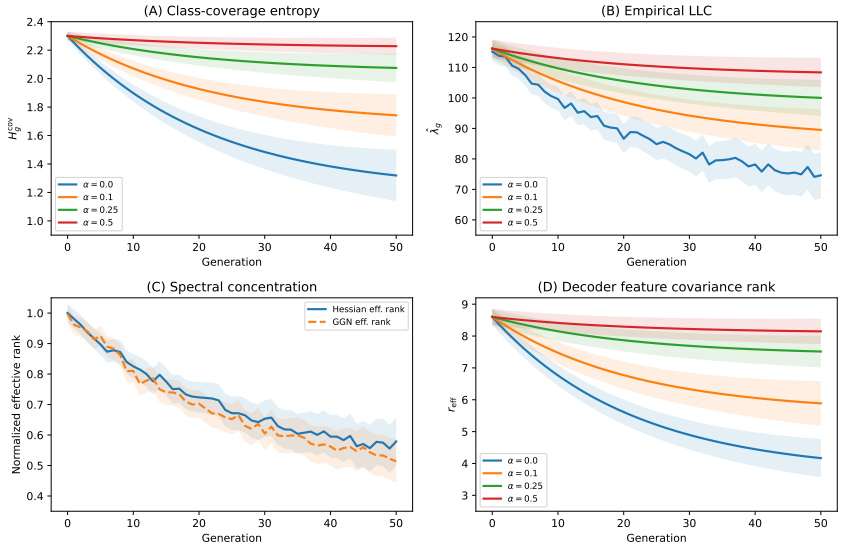


Figure 2: **VAE collapse signatures without explicit mixtures.** Under synthetic-only recursion, (A) decreasing class entropy indicates mode loss and (B) declining empirical LLC supports complexity decay across generations. (C) Hessian and Gauss–Newton effective ranks decline with spectral concentration, as does (D) decoder feature-covariance effective rank, which injection stabilizes. MNIST results use latent $d = 10$, $n = 5000$, and $G = 50$, with 10-seed means and 1σ bands.

3.4 Tier 3 — Standard VAEs: empirical signatures without explicit mixtures

A standard MNIST VAE tests geometric signatures without explicit mixtures. It uses a single Gaussian prior, $d = 10$, the same MLP encoder and decoder architecture as VaDE, and 50 continuation epochs per generation with $\alpha \in \{0, 0.1, 0.25, 0.5\}$. We track mode coverage and loss geometry rather than latent-component extinction.

Recursion lowers MNIST class entropy, decoder feature rank, and LLC, while Hessian and Gauss–Newton effective ranks roughly halve. Fixed-real training keeps entropy and feature rank near initialization, and injection at $\alpha \geq 0.25$ stabilizes all diagnostics at generation 50 (Figure 2, Table 1). The joint response of distribution and geometry supports (P2)–(P4) without the exact latent-prior nullity mechanism.

3.5 Tier 4 — DDPM: empirical signatures beyond latent-variable models

A compact 1.8M-parameter DDPM on FashionMNIST tests transfer beyond latent-variable models, using $\alpha \in \{0, 0.1, 0.25\}$. It has neither an explicit mixture prior nor a VAE-style latent decoder matching the final-layer identity, so the comparison concerns distributional degradation, feature and loss-geometry rank concentration, and stabilization (P3)–(P4) rather than latent-prior nullity (P1). We pair class entropy and FID with U-Net feature covariances and a blockwise empirical GGN matrix of the denoising loss.

Recursion lowers coverage and both ranks while raising FID. Fixed-real training keeps FID and ranks near initialization, and $\alpha = 0.25$ also preserves coverage (Figure 3, Table 1). The joint response of geometry and sample quality thus extends beyond VAE-style models.

3.6 Matched controls, full-model LLC, and autoregressive transfer

Matched controls separate recursive data replacement from continued optimization. Across VaDE, VAE, and the 12M CIFAR-10 DDPM, full-model LLC falls under recursion but stays near initialization with fixed-real training, and $\alpha = 0.25$ prevents most of the loss. Paired confidence intervals exclude zero, supporting attribution to data replacement (Table 13).

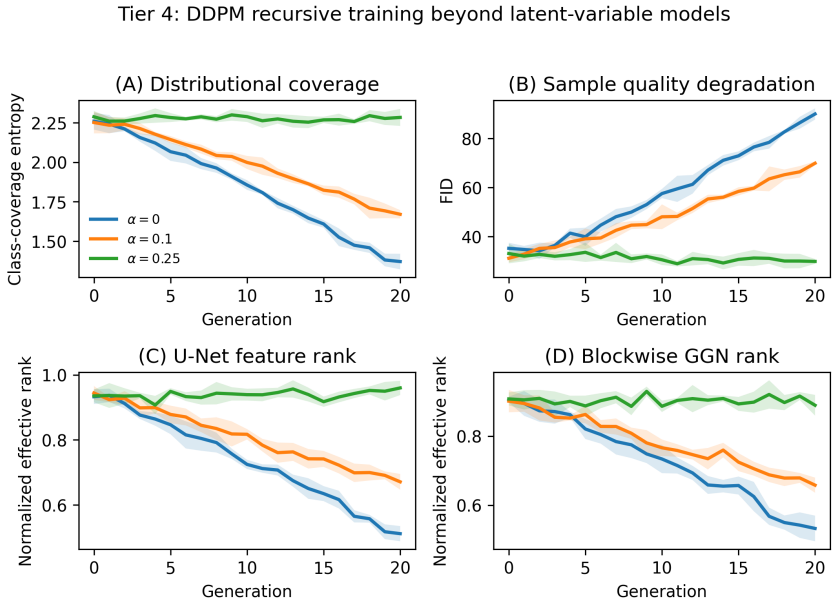


Figure 3: **DDPM geometry-collapse signatures beyond latent-variable models.** For a compact FashionMNIST DDPM with $\alpha \in \{0, 0.1, 0.25\}$, synthetic-only recursion ($\alpha = 0$) raises FID and lowers class entropy, U-Net feature-covariance effective rank, and blockwise Gauss–Newton effective rank. Injection at $\alpha = 0.25$ stabilizes these distributional and geometric diagnostics, largely preserving coverage, sample quality, and both ranks through $G = 20$. Results show 5-seed means with 1σ bands.

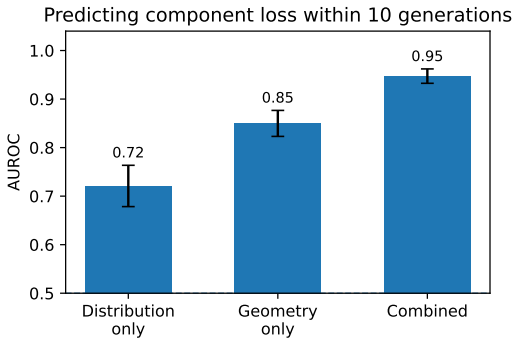


Figure 4: **Loss geometry predicts future collapse.** VaDE predictors forecast a decrease in active latent-component count r_g^{ε} after $\Delta = 10$ recursive generations, using leave-one-seed-out splits that keep trajectories intact. Geometry-only predictors outperform distribution-only ones, and combined predictors perform best. Bars show mean AUROC across held-out-seed folds with 1σ error bars.

Held-out performance follows the same pattern: recursion increases VaDE and VAE negative log-likelihood, whereas fixed-real training keeps both near initialization (Table 1). Complexity and predictive performance vary together, helping distinguish recursion from continued optimization.

Autoregressive transfer. OPT-125M tests this joint response in language on WikiText-2 across five seeds. Recursive LLC falls 38.9% as perplexity rises, coverage of the rarest token-frequency quintile falls, and hidden-state and GGN ranks decline. Fixed-real training preserves perplexity and rare-token coverage, while $\alpha = 0.25$ preserves these metrics and both ranks (Table 1, Appendix I).

Table 2: **Held-out-seed early-warning AUROC.** VaDE and VAE use MNIST, the compact DDPM uses FashionMNIST, and larger DDPMs use CIFAR-10.

Model	Distribution only	Geometry only	Combined
VaDE	0.72 ± 0.04	0.85 ± 0.03	0.95 ± 0.01
VAE	0.73 ± 0.04	0.84 ± 0.03	0.93 ± 0.02
DDPM-1.8M	0.70 ± 0.05	0.81 ± 0.04	0.90 ± 0.03
DDPM-12M	0.71 ± 0.04	0.86 ± 0.03	0.93 ± 0.02
DDPM-40M	0.68 ± 0.05	0.82 ± 0.04	0.89 ± 0.03
DDPM-96M	0.70 ± 0.04	0.84 ± 0.04	0.91 ± 0.03

Table 3: **Budget-matched prevention ($B = 0.10$).** Fixed injection uses $\alpha = 0.10$, while triggered policies use $\alpha_{\text{on}} = 0.25$. DDPM ranks are normalized, and variability is one seed standard deviation.

Model / metric	Fixed	Distribution	Random	Geometry
GMM / modes	9.4 ± 0.3	—	8.7 ± 0.5	9.9 ± 0.1
VaDE / components	8.2 ± 0.6	7.6 ± 0.7	6.3 ± 0.8	9.3 ± 0.4
VAE / classes	8.8 ± 0.5	8.6 ± 0.6	7.8 ± 0.7	9.6 ± 0.3
DDPM-1.8M / classes	8.6 ± 0.5	8.3 ± 0.6	7.5 ± 0.8	9.4 ± 0.4
VAE / feature rank	7.2 ± 0.5	6.9 ± 0.5	5.8 ± 0.6	8.0 ± 0.4
DDPM-1.8M / entropy	2.03 ± 0.06	1.96 ± 0.07	1.78 ± 0.09	2.19 ± 0.05
DDPM-1.8M / feature rank	0.81	0.76	0.64	0.90 ± 0.03
DDPM-1.8M / GGN rank	0.77	0.71	0.60	0.85 ± 0.04
DDPM-1.8M / FID	38.7 ± 3.0	40.5 ± 3.3	49.2 ± 4.1	32.8 ± 2.6

3.7 Prediction, intervention, and dose transfer

Early warning. We test whether geometry adds predictive information about support loss using logistic predictors with distributional metrics, geometric metrics, or both. VaDE targets endpoint loss, $\mathcal{C}_{g,\Delta} = \mathbf{1}\{r_{g+\Delta} < r_g\}$, whereas VAE and DDPM target loss within Δ generations (Appendix I.5). We use $\Delta = 10$ for VaDE and VAE, and 5 for DDPMs. VaDE distributional inputs comprise current active component count, class entropy, and held-out negative log-likelihood. Geometric inputs are componentwise latent-prior Fisher curvature, near-nullity, LLC, Hessian and GGN effective ranks, and decoder feature-covariance rank. Leave-one-seed-out evaluation keeps trajectories intact to test prediction on unseen seeds. Across VaDE and the other neural models through 96M parameters, geometry improves prediction, with combined inputs performing best (Figure 4, Table 2).

Cost at scale. Monitoring adds modest wall-clock overhead in the 12M, 40M, and 96M CIFAR-10 DDPMs, with proxies tracking sparse-checkpoint LLC closely. Block-local LLC at 40M and 96M also falls under recursion but stays near initialization with fixed-real training (Table 16).

Budget-matched prevention. We test whether predicted risk improves allocation of a shared injection budget. GMMs use $K = 10$, $G = 50$, $n = 5000$, and 100 held-out seeds, while VaDE and VAE use $G = 50$ and 10 held-out seeds. Compact DDPMs use $G = 20$ and five held-out seeds. A validation-frozen predictor triggers $\alpha_{\text{on}} = 0.25$ next generation when risk exceeds its threshold and budget remains. The GMM oracle uses minimum component prevalence. Coverage classifiers use only real MNIST (VAE) or FashionMNIST (DDPM) data (Appendix I.6).

At average real-data budget $B = 0.10$, geometric triggering preserves more modes, components, or classes than fixed $\alpha = 0.10$ and random triggering in all four models, and distribution triggering in all three neural models (Table 3). VAE feature rank and all four DDPM metrics also improve, with DDPM FID approaching fixed $\alpha = 0.25$ despite using 60% fewer real examples (Table 1).

Paired VaDE branches isolate intervention after a geometric alarm, sharing identical checkpoints and subsequent randomness, with alarms at least ten generations from all others. One intervention reduces ten-generation component loss from 62% to 14%.

Practical dose selection. We select the injection dose using the GMM scale $s = \log(K_{\text{eff}} G/\delta)/(n\widehat{\pi}_{\min})$, based on protected real-data classes or clusters and a conservative rare-

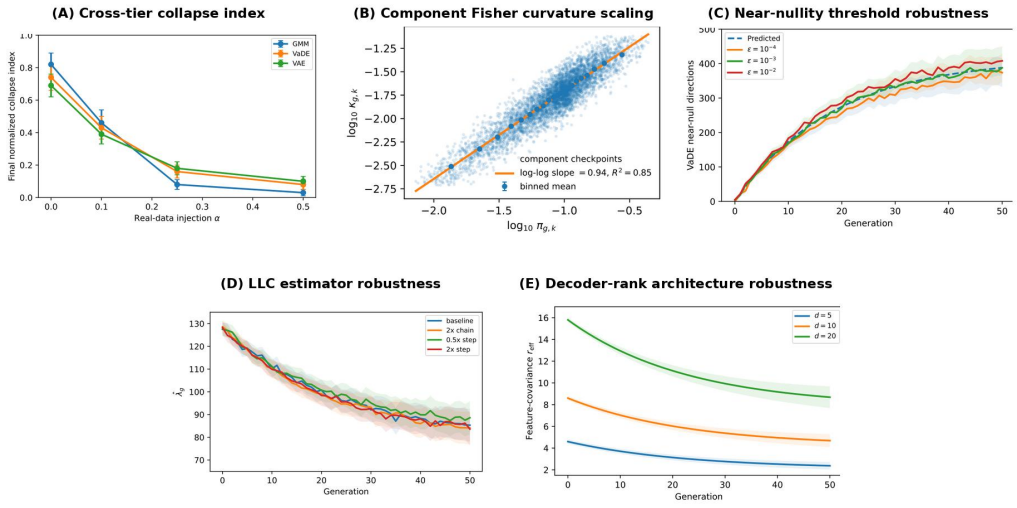


Figure 5: **Robustness of singular-geometric collapse diagnostics.** (A) More real-data injection α lowers the final normalized collapse index for GMM, VaDE, and standard VAE. (B) VaDE componentwise latent-prior Fisher curvature scales with mixture weight (log-log slope 0.94, $R^2 = 0.85$), consistent with the $O(\pi_{g,k})$ upper bound in Section 2.3. (C) VaDE latent-prior near-nullity remains stable across near-extinction thresholds ε, as does (D) synthetic-only LLC decay across SGLD estimator settings. (E) Decoder feature-covariance rank collapse persists across latent dimensions. All panels show 10-seed means with 1σ bands where applicable.

stratum prevalence estimate. Across 12 GMM and 12 neural settings, minimum protective doses closely follow this scale (Table 11), motivating feasible development choices near s. Frozen doses give similar protection to fixed $\alpha = 0.25$ on held-out VAE and DDPM with 40% fewer real examples.

Prospective protection protocol. Here, $\alpha_{\min}$ is the smallest tested dose retaining all protected classes or modes through G in at least 90% of held-out runs (Table 11). Using inputs computed from real data before recursion, we select one multiplicative calibration on GMM and VaDE development settings and freeze it for held-out VAE and DDPM without tuning individual test trajectories.

Real-data injection and robustness. In GMM, VaDE, and VAE, final normalized collapse indices decrease monotonically with α, with strong stabilization at $\alpha \geq 0.25$ (Figure 5A, Appendix I.2). DDPM shows the same stabilization beyond these models.

Robustness checks vary VaDE candidate near-nullity thresholds over $\varepsilon \in \{10^{-4}, 10^{-3}, 10^{-2}\}$ (Figure 5C), double SGLD chain length and perturb LLC step sizes (D), and repeat decoder feature-covariance measurements at latent $d \in \{5, 10, 20\}$ (E). These checks support empirical stability.

4 Discussion and limitations

This work reframes recursive model collapse as a loss-geometry phenomenon. In labeled mixtures, component extinction induces target-indexed RLCT descent and a growing Fisher-nullity lower bound; in VaDE, near-extinct components already have $O(\pi_k)$ Fisher-curvature upper bounds under bounded score moments. Standard VAEs and DDPMs lack this exact latent-prior mechanism, but show complexity decay, spectral concentration, feature-rank collapse, and stabilization under real-data injection. Matched controls and the OPT experiment strengthen the empirical comparison.

Real-data injection provides both a control and a finite-horizon labeled-mixture guarantee at the sufficient scale $\log(KG/\delta)/(n\pi^*_{\min})$. LLC, Hessian, and Gauss–Newton monitoring can be more expensive than standard sample-quality metrics. Multimodal models and broader settings remain directions for future work.

Acknowledgments and Disclosure of Funding

This work was self-funded by the author and received no third-party funding or support. The author declares no competing interests.

References

Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard Baraniuk. Self-consuming generative models go MAD. In *International Conference on Learning Representations (ICLR)*, 2024.

Miki Aoyagi. A Bayesian learning coefficient of generalization error and Vandermonde matrix-type singularities. *Communications in Statistics — Theory and Methods*, 39(15):2667–2687, 2010.

Miki Aoyagi and Kenji Nagata. Learning coefficient of generalization error in Bayesian estimation and Vandermonde matrix-type singularity. *Neural Computation*, 24(6):1569–1610, 2012.

Matteo Benati, Alessandro Londei, Denise Lanzieri, and Vittorio Loreto. First-extinction law for resampling processes. *arXiv preprint arXiv:2509.20101*, 2025.

Quentin Bertrand, Avishek Joey Bose, Alexandre Duplessis, Marco Jiralerspong, and Gauthier Gidel. On the stability of iterative retraining of generative models on their own data. In *International Conference on Learning Representations (ICLR)*, 2024.

Vivek Shripad Borkar. A theoretical basis for model collapse in recursive training. *arXiv preprint arXiv:2506.09401*, 2025.

Zhongtian Chen, Edmund Lau, Jake Mendel, Susan Wei, and Daniel Murfet. Dynamical versus Bayesian phase transitions in a toy model of superposition. *arXiv preprint arXiv:2310.06301*, 2023.

Elvis Dohmatob, Yunzhen Feng, and Julia Kempe. Model collapse demystified: The case of regression. *arXiv preprint arXiv:2402.07712*, 2024.

Warren J. Ewens. *Mathematical Population Genetics: I. Theoretical Introduction*. Springer, 2 edition, 2004.

Luca Ferretti and Paolo Ribeca. Allelic entropy and times until loss or fixation of neutral polymorphisms. *Genes*, 14(9):1751, 2023. doi: 10.3390/genes14091751.

Zach Furman and Edmund Lau. Estimating the local learning coefficient at scale. *arXiv preprint arXiv:2402.03698*, 2024.

Ryuichiro Hataya, Han Bao, and Hiromi Arai. Will large-scale generative models corrupt future datasets? In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Jesse Hoogland, George Wang, Matthew Farrugia-Roberts, Liam Carroll, Susan Wei, and Daniel Murfet. The developmental landscape of in-context learning. *arXiv preprint arXiv:2402.02364*, 2024.

Fariba Jangjoo, Matteo Marsili, and Yasser Roudi. Lost in retraining: Roaming the parameter space of exponential families under closed-loop learning. *arXiv preprint arXiv:2506.20623*, 2025.

Zhuxi Jiang, Yin Zheng, Huachun Tan, Bangsheng Tang, and Hanning Zhou. Variational deep embedding: An unsupervised and generative approach to clustering. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1965–1972, 2017.

Motoo Kimura. Stochastic processes and distribution of gene frequencies under natural selection. *Cold Spring Harbor Symposia on Quantitative Biology*, 20:33–53, 1955.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. URL `https://www.cs.toronto.edu/~kriz/cifar.html`.

Yuki Kurumadani. Learning coefficients in semiregular models ii: extensions. *Japanese Journal of Statistics and Data Science*, 9(2):249–283, 2026. doi: 10.1007/s42081-025-00310-4. URL https://doi.org/10.1007/s42081-025-00310-4. Published online 31 August 2025.

Edmund Lau, Zach Furman, George Wang, Daniel Murfet, and Susan Wei. The local learning coefficient: A singularity-aware complexity measure. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 244–252, 2025.

Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791. URL https://bottou.org/papers/lecun-98h.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models, 2016. URL https://arxiv.org/abs/1609.07843.

Søren Riis. Drift and selection in LLM text ecosystems. *arXiv preprint arXiv:2604.08554*, 2026.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. AI models collapse when trained on recursively generated data. *Nature*, 631(8022):755–759, 2024. doi: 10.1038/s41586-024-07566-y.

Ananda Theertha Suresh, Andrew Thangaraj, and Aditya Nanda Kishore Khandavally. Rate of model collapse in recursive training. *arXiv preprint arXiv:2412.17646*, 2024.

D. M. Titterington, A. F. M. Smith, and U. E. Makov. *Statistical Analysis of Finite Mixture Distributions*. Wiley, Chichester, 1985.

George Wang, Jesse Hoogland, Stan van Wingerden, Zach Furman, and Daniel Murfet. Differentiation and specialization of attention heads via the refined local learning coefficient. *arXiv preprint arXiv:2410.02984*, 2024.

Sumio Watanabe. *Algebraic Geometry and Statistical Learning Theory*. Cambridge University Press, 2009.

Sumio Watanabe. *Mathematical Theory of Bayesian Statistics*. Chapman and Hall/CRC, 2018.

Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017. URL https://arxiv.org/abs/1708.07747.

Keisuke Yamazaki, Miki Aoyagi, and Sumio Watanabe. Asymptotic analysis of Bayesian generalization error with Newton diagram. *Neural Networks*, 23(1):35–43, 2010.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, Todor Mihaylov, Myle Ott, Sam Shleifer, Kurt Shuster, Daniel Simig, Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. OPT: Open pre-trained transformer language models, 2022. URL https://arxiv.org/abs/2205.01068.

A Related work

Model collapse in recursive generative training. Shumailov et al. [2024] demonstrated that recursive training of generative models on their own outputs causes progressive distributional degradation, with rare modes lost first and tail mass disappearing across generations. Subsequent work has established this phenomenon across model classes: Alemohammad et al. [2024] for self-consuming generative loops, Hataya et al. [2023] for image-generation pipelines, Bertrand et al. [2024] for iterative retraining stability, and Dohmatob et al. [2024] for the regression case. These accounts characterize collapse *distributionally*, through variance shrinkage, tail loss, and divergence metrics. A second line of work analyzes the underlying *probabilistic dynamics*: Suresh et al. [2024] establish first-extinction rates for $K = 2$ multinomial recursion, Borkar [2025] prove vertex absorption for general categorical chains, Jangjoo et al. [2025] treat the Random Energy Model case, Benati et al. [2025] derive Wright–Fisher first-extinction laws, and Riis [2026] observe that the same dynamics describe recursive n-gram language model retraining. Our work builds on this dynamical foundation but addresses a different question: not what extinction does to the data distribution or the mixture-weight chain, but what it does to the singular geometry of the parameter space. The closed-form RLCT trajectory of Theorem 1 characterizes a complexity quantity that no prior collapse paper computes.

Singular learning theory and the local learning coefficient. Singular learning theory [Watanabe, 2009, 2018] provides the framework in which the real log-canonical threshold quantifies effective parameter complexity for singular models, including neural networks. Aoyagi [2010] and Aoyagi and Nagata [2012] studied normal-mixture learning coefficients via Vandermonde matrix-type singularities. For nonnegative unit-variance Gaussian mixtures, we use the coefficient reported by Kurumadani [2026] and proved in Appendix D. Related Newton-diagram calculations for binomial mixtures appear in Yamazaki et al. [2010]. The local learning coefficient (LLC) of Lau et al. [2025], with the scalable estimator of Furman and Lau [2024], provides a local empirical complexity diagnostic for deep networks via stochastic-gradient Langevin dynamics. Recent applications include Hoogland et al. [2024] on developmental phase transitions in transformer training, Chen et al. [2023] on toy models of superposition, and Wang et al. [2024] on attention-head specialization. To our knowledge, no prior work has applied SLT to recursive generative training, nor connected the LLC to model-collapse dynamics. Our Theorem 1 provides the first closed-form RLCT trajectory along a recursive-training process; the results in Section 2.3 extend the unidentifiability mechanism to neural latent-variable models.

Population-genetics absorption times. Neutral Wright–Fisher dynamics provide the classical comparison for multinomial fixation times [Kimura, 1955, Ewens, 2004, Ferretti and Ribeca, 2023]. Recursive-training connections appear in Benati et al. [2025], Riis [2026].

A.1 Scope of the mixture learning coefficient

The recursion retains component labels for its update, whereas its geometric quantities refer to the marginal density $p_\theta(x)$. For a target q, write $K_q(\theta) = \mathrm{KL}(q\|p_\theta)$ and $\mathcal{Z}(q) = \{\theta : p_\theta = q\}$. The coefficient in Section 2.2 is the model-level coefficient

$$\lambda(K, r) = \min_{\theta \in \mathcal{Z}(q)} \lambda_\theta(K_q),$$

for a target with r distinct positive-weight components, under the explicit conditions of Theorem 2. Local coefficients use nonnegative simplex weights and a smooth density bounded above and below by positive constants relative to affine-weight and mean Lebesgue measure, including boundary strata. For each finite-generation target, a compact parameter region and prior support must include a representation attaining the minimum. This target-wise convention avoids imposing a fixed compact mean domain on the unbounded Gaussian recursion. The closed form does not apply without qualification when a prior vanishes on the contributing strata or its support excludes them.

This minimum need not equal the local coefficient at the particular checkpoint θ_g. For example, let $K = 2$, $q = \mathcal{N}(0, 1)$, and consider $\pi = (1, 0)$, $\eta = (0, b)$ with $b \neq 0$. For the active-mean and inactive-weight coordinates, the marginal scores are

$$x, \qquad e^{bx - b^2/2} - 1.$$

Their Fisher Gram determinant is $e^{b^2} - 1 - b^2 > 0$. Consequently the local KL is comparable to $u^2 + a^2$, with $a \geq 0$, uniformly for inactive means near b; its local coefficient is 1. The model-level value is instead $\lambda(2, 1) = 3/4$, attained at merged representations. The same formula also need not hold for the joint likelihood of observed component labels.

The component count r_g is the number of positive-weight components with distinct means, not necessarily the number of local maxima of the density. Conditional on positive counts, the updated sample means have continuous joint densities, so any two surviving means coincide with probability zero. A countable union over generations preserves distinctness almost surely; no uniform separation margin or multimodality claim follows. The same argument applies to the Gaussian mixed-sample update on the no-extinction event.

For the Fisher matrix, gradients use $K - 1$ affine simplex coordinates and the K mean coordinates, with boundary scores interpreted by their one-sided limits. An extinct component's mean is retained at its previous value, making θ_g a defined representative without affecting the density. Its mean score vanishes, yielding the $|E_g|$ independent null directions used in the theorem. Fisher rank is invariant under nonsingular coordinate changes, while the RLCT additionally requires the corresponding prior measure (or comparable positive local densities); a finite softmax chart does not cover the simplex boundary.

B Proof of the RLCT trajectory theorem

Proof of Theorem 1. Conditional on positive counts, surviving Gaussian sample means have continuous densities; pairwise coincidences have probability zero. A countable union over generations preserves distinct active means almost surely. In the labeled recursion, if component k is extinct at generation g, then $\pi_{g,k} = 0$. Hence $N_{g,k} = 0$ almost surely at the next generation and $\pi_{g+1,k} = 0$. Therefore $E_g \subseteq E_{g+1}$ and $r_{g+1} \leq r_g$.

By the coefficient proved in Appendix D, $\lambda(K, r)$ is strictly increasing in the number r of distinct positive-weight components, with

$$\lambda(K, r) - \lambda(K, r - 1) = \frac{3}{4}.$$

Since $\lambda_g = \lambda(K, r_g)$ and r_g is non-increasing, λ_g is non-increasing almost surely. If $d = r_g - r_{g+1}$ components become extinct, the RLCT drops by exactly $3d/4$.

The weight process is the neutral finite-population Wright–Fisher chain

$$\pi_{g+1} = N_g/n, \qquad N_g \sim \text{Multinomial}(n, \pi_g).$$

Its absorbing states are the simplex vertices. Starting from strictly positive weights, the chain is absorbed at a vertex almost surely, so eventually exactly one component survives. Hence $r_g \to 1$ and

$$\lambda_g \to \lambda(K, 1).$$

Finally, if $k \in E_g$, then the model density at generation g no longer depends on the component mean η_k. Therefore

$$\partial_{\eta_k} \log p_{\theta_g}(x) = 0$$

for all x in the support of the surviving mixture, so the η_k-coordinate is a Fisher-null direction. Thus

$$\dim \ker I(\theta_g) \geq |E_g|.$$

Since the Wright–Fisher chain is eventually absorbed at a single surviving component, $|E_g| \to K - 1$, and therefore

$$\liminf_{g \to \infty} \dim \ker I(\theta_g) \geq K - 1.$$

□

C Setup and notation

Univariate Gaussian mixture model. For the main theorem we analyze a K-component univariate Gaussian mixture with known unit variance:

$$p_\theta(x) \;=\; \sum_{k=1}^{K} \pi_k \, \phi(x \mid \eta_k), \qquad \phi(x \mid \eta) := \tfrac{1}{\sqrt{2\pi}} \exp\!\big(-\tfrac{1}{2}(x - \eta)^2\big), \tag{1}$$

parameterized by mixture weights $\pi \in \Delta^{K-1} := \{\pi \in \mathbb{R}^K_{\geq 0} : \sum_k \pi_k = 1\}$ and component means $\eta_k \in \mathbb{R}$. The full parameter is $\theta = (\pi, \eta_1, \ldots, \eta_K) \in \Theta := \Delta^{K-1} \times \mathbb{R}^K$, with $D := \dim \Theta = 2K - 1$. The data-generating parameter is $\theta^* = (\pi^*, \eta_1^*, \ldots, \eta_K^*)$ with $p^* := p_{\theta^*}$. Minimal positive-weight representations with distinct means are identifiable up to component relabeling [Titterington et al., 1985]; we work on the quotient $\Theta/\mathfrak{S}_K$, while retaining the separate nonidentifiability caused by zero weights and coincident means.

Remark 1 (General exponential-family extension). The coefficient in Theorem 2 applies to scalar Gaussian means with known common variance one and ordinary nonnegative simplex weights. The same numerical formula is not asserted for higher-dimensional or general exponential-family mixtures.

Assumption 2 (Non-degenerate data generator)**.** *$\pi_k^* > 0$ for all k and $\eta_j^* \neq \eta_k^*$ for all $j \neq k$.*

Recursive synthetic-only training (labeled abstraction). Our theoretical analysis treats a *labeled* (oracle-separated) recursion, in which generated samples retain their latent component identity. This abstraction isolates the singular-geometric mechanism from the unlabeled-mixture identifiability problem, and matches the abstraction used implicitly by Suresh et al. [2024], Borkar [2025], Jangjoo et al. [2025], Benati et al. [2025] in their analyses of recursive multinomial dynamics. The neural experiments of Section 3 test whether the same singular-geometric signatures persist in unlabeled, end-to-end-trained generative models.

Fix sample size $n \in \mathbb{N}$ per generation. Starting from $\theta_0 = \theta^*$, define $\{\theta_g\}_{g \geq 0}$ as follows. At generation g, draw labeled synthetic samples

$$(x_i^{(g)}, z_i^{(g)})_{i=1}^n \overset{\text{i.i.d.}}{\sim} p_{\theta_g}, \qquad z_i^{(g)} \in \{1, \ldots, K\}, \tag{2}$$

where $z_i^{(g)} \mid \theta_g \sim \mathrm{Categorical}(\pi_g)$ is the latent component identity and $x_i^{(g)} \mid z_i^{(g)} = k, \theta_g \sim \phi(\cdot \mid \eta_{g,k})$. Let $N_{g,k} := |\{i : z_i^{(g)} = k\}|$ denote the count of samples drawn from component k, so that $(N_{g,1}, \ldots, N_{g,K}) \mid \theta_g \sim \mathrm{Multinomial}(n, \pi_g)$. The next-generation parameter is the labeled-MLE update

$$\pi_{g+1,k} \;=\; \frac{N_{g,k}}{n}, \qquad \eta_{g+1,k} \;=\; \frac{1}{N_{g,k}} \sum_{i : z_i^{(g)} = k} x_i^{(g)} \quad \text{whenever } N_{g,k} \geq 1, \tag{3}$$

When $N_{g,k} = 0$, retain $\eta_{g+1,k} = \eta_{g,k}$ so that the full parameter remains defined; this coordinate does not affect the current density. Let $\mathcal{F}_g = \sigma(\{(x_i^{(j)}, z_i^{(j)}) : 1 \leq i \leq n,\ 0 \leq j < g\})$ be the natural filtration.

Extinct set. For generation g, define the *extinct set* $E_g := \{k \in \{1, \ldots, K\} : \pi_{g,k} = 0\}$ and the *support* $S_g := \{1, \ldots, K\} \setminus E_g$. We say component k is *extinct at generation g* if $k \in E_g$ and *surviving* otherwise.

Fisher information and loss-landscape diagnostics. The Fisher information at $\theta \in \Theta$ is

$$I(\theta) \;=\; \mathbb{E}_{x \sim p_\theta}\left[\nabla_\theta \log p_\theta(x)\, \nabla_\theta \log p_\theta(x)^\top\right] \in \mathbb{R}^{D \times D}. \tag{4}$$

Mixture models are singular on the boundary stratum $\partial_{\deg}\Theta := \{\theta : \min_k \pi_k = 0\}$ [Watanabe, 2009, Aoyagi, 2010]. For sample $X = (x_1, \ldots, x_m)$ we write $\mathcal{L}(\theta; X) := -m^{-1} \sum_i \log p_\theta(x_i)$. The diagnostics at θ_g are: the empirical Hessian $H_g := \nabla^2 \mathcal{L}(\theta_g; X^{(g)})$ and its spectrum; the Gauss–Newton matrix G_g (defined when used); the model-level real log-canonical threshold λ_g [Watanabe, 2009]; and the separate empirical local learning coefficient $\hat{\lambda}_g$ of Lau et al. [2025], computed via SGLD as in Furman and Lau [2024]. Hessian spectra and the RLCT are not equivalent for singular models; we treat them as complementary.

Real-data injection. The α-mixed recursion replaces $X^{(g)}$ in (2) with $\lfloor (1-\alpha)n \rfloor$ labeled samples from p_{θ_g} and $\lceil \alpha n \rceil$ labeled samples from p^*, drawn independently; $\alpha = 0$ is synthetic-only and $\alpha = 1$ is standard MLE from p^*.

C.1 Background: recursive multinomial dynamics

After the first update, the mixture-weight component of the recursion (3) is a finite-state Markov chain on the rational simplex $\Delta_n^{K-1} := \{\pi \in \Delta^{K-1} : n\pi_k \in \mathbb{N}_0\}$. The following facts are established in the recent recursive-training literature; we record them here as background and use them as scaffolding for the main results in the next subsection.

Lemma 1 (Martingale property)**.** *Each coordinate $\{\pi_{g,k}\}_{g\geq 0}$ is a bounded $\mathcal{F}_g$-martingale: $\mathbb{E}[\pi_{g+1,k} \mid \mathcal{F}_g] = \pi_{g,k}$.*

Lemma 2 (Vertex absorption)**.** *The simplex vertices $\{e_1, \ldots, e_K\}$ are the only absorbing states of the chain $\{\pi_g\}$. With probability one, there exists a random index $J \in \{1, \ldots, K\}$ such that $\pi_g \to e_J$, with absorption probabilities*

$$\mathbb{P}(J = k) \;=\; \pi_{0,k}, \qquad k = 1, \ldots, K.$$

The expected first-extinction time is finite for every $K \geq 2$ and every $n \geq 1$.

Lemma 3 (Monotone extinction)**.** *The extinct set is non-decreasing: $E_g \subseteq E_{g+1}$ almost surely. Equivalently, once $\pi_{g,k} = 0$ for some g, then $\pi_{g',k} = 0$ for all $g' \geq g$.*

Proof. The conditional multinomial mean gives $\mathbb{E}[N_{g,k} \mid \mathcal{F}_g] = n\pi_{g,k}$, proving the martingale property. At any state, some weight is at least $1/K$, so the next generation is a vertex with conditional probability at least K^{-n}. Hence the absorption time T satisfies $\mathbb{P}(T > m) \leq (1 - K^{-n})^m$ and $\mathbb{E}[T] \leq K^n < \infty$. Nonvertex states are not absorbing, while vertices are. Bounded convergence in the coordinate martingales gives $\mathbb{P}(J = k) = \mathbb{E}[\pi_{T,k}] = \pi_{0,k}$. First extinction occurs no later than absorption. Finally, $\pi_{g,k} = 0$ implies $N_{g,k} = 0$ almost surely, proving monotone extinction. □

Lemma 1 appears in essentially this form in Borkar [2025] and Jangjoo et al. [2025], the latter as a consequence of a more general martingale property for sufficient statistics of exponential families under closed-loop learning. Lemma 2 is established for the $K = 2$ case by Suresh et al. [2024] and for the categorical / Random Energy Model case by Jangjoo et al. [2025]; Benati et al. [2025] prove a closely related first-extinction law via Wright–Fisher diffusion. Riis [2026] observes that the same dynamics describe recursive n-gram language model retraining. Lemma 3 is implicit in all of these but, to our knowledge, is not stated as a separate result; we record it explicitly because it is what drives the singularity statement of the next subsection.

Remark 2. The surviving means are updated from finite random sub-samples; fixed-size recursive updates do not imply consistency for their initial values. At a finite generation with distinct surviving means and positive surviving weights, the surviving sub-model is locally regular. The geometric phenomenon we identify — progressive degeneracy of the full-model Fisher information along the trajectory — is a consequence of *which directions* in Θ become unidentifiable as components go extinct, not of the absorption itself.

D Learning coefficient of a nonnegative Gaussian location mixture

The nonnegativity of mixture weights is essential to the calculation below. For a univariate Gaussian mixture with known variance one, the coefficient is $(K + 3r - 2)/4$. Example 5.3 of Kurumadani [2026] gives this value and states that its bound is tight for variance-one normal mixtures. We give a self-contained proof, including simplex-boundary representations.

Parameter domain and prior. Fix integers $1 \leq r \leq K$ and a nondegenerate compact interval $I = [L, R] \subset \mathbb{R}$. Put

$$\phi_\eta(x) = (2\pi)^{-1/2} \exp\{-\tfrac{1}{2}(x - \eta)^2\}, \qquad p_\theta(x) = \sum_{i=1}^{K} a_i \phi_{\eta_i}(x), \qquad \Theta_I = \Delta^{K-1} \times I^K,$$

where $a_i \geq 0$ and $\sum_i a_i = 1$. The target is

$$q(x) = \sum_{\ell=1}^{r} \rho_\ell \phi_{\mu_\ell}(x), \qquad \rho_\ell > 0, \quad \sum_\ell \rho_\ell = 1,$$

with pairwise distinct $\mu_\ell \in (L, R)$. The prior has a smooth, strictly positive density φ up to the boundary of Θ_I, relative to Lebesgue measure on the affine hyperplane $\sum_i a_i = 1$ and on I^K. Equivalently, one may use any $K-1$ independent weight coordinates; the constant change in volume measure does not affect the result. It suffices throughout that the density be locally bounded above and below by positive constants. A prior singular or vanishing at a simplex face is not covered by this statement.

Write

$$\mathcal{K}_q(\theta) = \int_{\mathbb{R}} q(x) \log \frac{q(x)}{p_\theta(x)}\, dx, \qquad \mathcal{Z}_q = \{\theta \in \Theta_I : p_\theta = q\}.$$

At $\theta^0 \in \mathcal{Z}_q$, define the local learning coefficient as the integrability threshold of $\mathcal{K}_q^{-s}\varphi$ on a sufficiently small relative neighborhood of θ^0. The global coefficient is the same threshold on Θ_I. This definition does not assume that the Gaussian mixture KL divergence is analytic. The proof establishes the threshold by two-sided comparison with explicit polynomial functions and also establishes the corresponding power law for the local and global Laplace integrals.

Theorem 2 (Mixture coefficient, including boundary strata)**.** *Under the preceding conditions, let $e(\theta^0)$ be the number of indices i such that $a_i^0 = 0$ and $\eta_i^0 \notin \{\mu_1, \dots, \mu_r\}$. Then*

$$\lambda_{\theta^0} = \frac{K + 3r - 2 + e(\theta^0)}{4}.$$

Consequently,

$$\boxed{\lambda(K, r) = \min_{\theta^0 \in \mathcal{Z}_q} \lambda_{\theta^0} = \frac{K + 3r - 2}{4}.}$$

In particular,

$$\lambda(K, K) = \frac{2K - 1}{2}, \qquad \lambda(K, r) - \lambda(K, r-1) = \frac{3}{4} \quad (2 \le r \le K).$$

For each sufficiently small local neighborhood U, and globally on the compact domain, there are positive constants bounding the corresponding Laplace integral above and below by $t^{-\lambda}$ for all sufficiently large t. In particular, there is no additional logarithmic factor in these two-sided estimates.

Proof. We use $f \asymp g$ to mean comparison in both directions by positive constants on the neighborhood under discussion. Constants may depend on the target and the base representation; no uniform separation assumption over a recursive trajectory is being made.

1. Classification of a target representation. Fix $\theta^0 \in \mathcal{Z}_q$. Gaussian translates at distinct means are linearly independent: dividing a putative linear relation by $\exp(-x^2/2)$ gives a relation among exponentials with distinct real exponents. Thus every positive-weight atom of θ^0 has a true mean, and the total weight at μ_ℓ is ρ_ℓ.

Let $I_\ell = \{i : \eta_i^0 = \mu_\ell\}$, and put $s_\ell = |I_\ell|$. Each I_ℓ contains a positive-weight pivot i_ℓ. The remaining $e = K - \sum_\ell s_\ell$ atoms all have zero weight. Group these remaining atoms into sets J_h according to their distinct base means ν_h, $1 \le h \le H$. The ν_h differ from all true means; several inactive atoms are allowed to have the same ν_h. When there are no such atoms, sums over h or the sets J_h are empty.

For $i \in I_\ell$ set $x_i = \eta_i - \mu_\ell$, and define

$$W_\ell = \sum_{i \in I_\ell} a_i, \quad U_\ell = W_\ell - \rho_\ell, \quad M_{1,\ell} = \sum_{i \in I_\ell} a_i x_i, \quad M_{2,\ell} = \sum_{i \in I_\ell} a_i x_i^2.$$

For $j \in J_h$ write $b_j = a_j$ and $B_h = \sum_{j \in J_h} b_j$. All $M_{2,\ell}$, b_j, and B_h are nonnegative, and

$$U_r = -\sum_{\ell < r} U_\ell - \sum_h B_h.$$

Shrink the neighborhood so that each pivot weight and each W_ℓ is bounded below by a positive constant.

2. A polynomial comparison for KL. Taylor expansion in the mean gives

$$p_\theta - q = \sum_{\ell<r} U_\ell(\phi_{\mu_\ell} - \phi_{\mu_r}) + \sum_{\ell=1}^{r} M_{1,\ell}\phi'_{\mu_\ell} + \frac{1}{2}\sum_{\ell=1}^{r} M_{2,\ell}\phi''_{\mu_\ell} + \sum_{h=1}^{H} B_h(\phi_{\nu_h} - \phi_{\mu_r}) + \mathcal{R},$$

where primes denote derivatives with respect to the mean. If all mean displacements from their base values have absolute value at most ϵ, then

$$\|\mathcal{R}\|_2 \le C\epsilon\left(\sum_\ell M_{2,\ell} + \sum_h B_h\right).$$

Indeed, a true-cluster remainder is bounded by a constant times $\sum_i a_i|x_i|^3 \le \epsilon M_{2,\ell}$; a foreign-cluster remainder is bounded by a constant times $\sum_{j\in J_h} b_j|\eta_j - \nu_h| \le \epsilon B_h$. These estimates are valid in $L^2(\mathbb{R})$ by the uniform bounds for Gaussian mean derivatives on a bounded mean interval.

The functions multiplying the displayed coefficients are linearly independent. To see this, restore the coefficient of ϕ_{μ_r} and divide by $\exp(-x^2/2)$. The resulting functions are polynomials of degree at most two times exponentials with distinct exponents. Such exponential polynomials are linearly independent; at each true mean the degrees zero, one, and two also have independent leading coefficients. Therefore the Gram matrix of the displayed finite family is positive definite. After decreasing ϵ, its smallest eigenvalue absorbs the remainder and gives

$$\|p_\theta - q\|_2^2 \asymp F_0, \qquad F_0 := \sum_{\ell<r} U_\ell^2 + \sum_\ell M_{1,\ell}^2 + \sum_\ell M_{2,\ell}^2 + \sum_{j\notin\cup_\ell I_\ell} b_j^2.$$

Here we used $\sum_{j\in J_h} b_j^2 \le B_h^2 \le |J_h|\sum_{j\in J_h} b_j^2$. This is the step that handles coincident foreign means and uses nonnegativity of the weights.

For completeness, the same comparison holds for KL without any analyticity assumption. Both p_θ and q are bounded by $M = (2\pi)^{-1/2}$. The elementary Hellinger and chi-square inequalities give

$$\mathcal{K}_q(\theta) \ge \int(\sqrt{q} - \sqrt{p_\theta})^2\,dx \ge \frac{1}{4M}\|p_\theta - q\|_2^2, \qquad \mathcal{K}_q(\theta) \le \int \frac{(p_\theta - q)^2}{p_\theta}\,dx.$$

To bound the latter integral uniformly, choose the positive pivot in the first true cluster. Fix a preliminary neighborhood radius $\epsilon_0 > 0$ and set

$$d_0(x) = \phi_{\mu_1}(x)\exp\{-\epsilon_0|x - \mu_1| - \epsilon_0^2/2\}.$$

For $|\eta_{i_1} - \mu_1| \le \epsilon_0$, $\phi_{\eta_{i_1}} \ge d_0$, so $p_\theta \ge cd_0$ with $c > 0$. Gaussian mean derivatives of each fixed order have uniformly finite norm in $L^2(d_0^{-1}\,dx)$ when their means range over a bounded interval. The same Taylor expansion consequently bounds $\|p_\theta - q\|^2_{L^2(d_0^{-1}\,dx)}$ by CF_0. We conclude that

$$\mathcal{K}_q(\theta) \asymp F_0. \tag{5}$$

No uniform bound on p_θ/q in the Gaussian tails has been used.

3. Coordinates and the simplex boundary. In a true cluster define its center displacement and its internal variance by

$$c_\ell = \frac{M_{1,\ell}}{W_\ell}, \qquad V_\ell = \sum_{i\in I_\ell} a_i(x_i - c_\ell)^2.$$

Then $M_{1,\ell} = W_\ell c_\ell$ and $M_{2,\ell} = W_\ell c_\ell^2 + V_\ell$. Since W_ℓ is bounded above and away from zero, c_ℓ is small, and $V_\ell \ge 0$,

$$M_{1,\ell}^2 + M_{2,\ell}^2 \asymp c_\ell^2 + V_\ell^2.$$

For each nonpivot $i \in I_\ell$ set $\delta_i = \eta_i - \eta_{i_\ell}$. A direct calculation yields

$$V_\ell = \sum_{i\ne i_\ell} a_i\delta_i^2 - \frac{1}{W_\ell}\left(\sum_{i\ne i_\ell} a_i\delta_i\right)^2,$$

where sums in this display are restricted to I_ℓ. Cauchy–Schwarz gives

$$\frac{a_{i_\ell}}{W_\ell}\sum_{i\neq i_\ell} a_i\delta_i^2 \le V_\ell \le \sum_{i\neq i_\ell} a_i\delta_i^2.$$

Thus $V_\ell \asymp \sum_{i\neq i_\ell} a_i\delta_i^2$.

Use as coordinates $U_1,\dots,U_{r-1}, c_1,\dots,c_r$, all nonpivot weights and differences within true clusters, and all foreign weights and mean displacements. The inverse map is explicit:

$$W_\ell = \rho_\ell + U_\ell \quad (\ell < r), \qquad W_r = \rho_r - \sum_{\ell<r} U_\ell - \sum_j b_j,$$

$$a_{i_\ell} = W_\ell - \sum_{i\in I_\ell\setminus\{i_\ell\}} a_i,$$

$$\eta_{i_\ell} = \mu_\ell + c_\ell - \frac{1}{W_\ell}\sum_{i\in I_\ell\setminus\{i_\ell\}} a_i\delta_i, \qquad \eta_i = \eta_{i_\ell} + \delta_i.$$

This is an analytic coordinate change with nonzero Jacobian near the base point. Its prior-weighted Jacobian is bounded above and below by positive constants.

In these coordinates a sufficiently small relative neighborhood is sandwiched between product neighborhoods. Variables whose base weights vanish range over one-sided intervals $[0, A)$; positive weights range over intervals about their base values. Pivot weights remain strictly positive by shrinking the boxes. True means lie in the interior of I, so the center and internal mean coordinates have two-sided intervals. A foreign mean at an endpoint of I has a one-sided interval, but is only a nuisance coordinate in the calculation below. Thus no extension to negative mixture weights, nor any removal of simplex-boundary neighborhoods, is implicit in the coordinate argument.

Split the nonpivot indices of the true clusters into P, with $a_i^0 > 0$, and Z, with $a_i^0 = 0$. There are $|P| + |Z| = K - e - r$ such indices. For $i \in P$, a_i is bounded above and away from zero. The terms in each sum defining V_ℓ are nonnegative; the square of their sum is therefore comparable to the sum of their squares. Equation (5) becomes

$$\mathcal{K}_q(\theta) \asymp F := \sum_{\ell<r} U_\ell^2 + \sum_{\ell=1}^{r} c_\ell^2 + \sum_{j \text{ foreign}} b_j^2 + \sum_{i\in P}\delta_i^4 + \sum_{i\in Z} a_i^2\delta_i^4. \tag{6}$$

All displayed summands use disjoint variables. The remaining coordinates are nuisance coordinates integrated over intervals of positive length.

4. Integration of the monomial comparison. An ordinary quadratic variable contributes a factor comparable to $t^{-1/2}$ to $\int e^{-tF}$, including a nonnegative quadratic variable on a one-sided interval. An ordinary quartic variable contributes $t^{-1/4}$. The remaining factor is, for fixed $A, D > 0$,

$$J(t) = \int_0^A \int_{-D}^{D} \exp(-ta^2\delta^4)\, d\delta\, da.$$

Set $v = \sqrt{a}\,\delta$ for $a > 0$ and integrate out a. This gives the exact identity

$$J(t) = \int_{-D\sqrt{A}}^{D\sqrt{A}} e^{-tv^4}\left(\int_{v^2/D^2}^{A} a^{-1/2}\, da\right) dv = \int_{-D\sqrt{A}}^{D\sqrt{A}} e^{-tv^4} 2\left(\sqrt{A} - \frac{|v|}{D}\right) dv.$$

The integrated density is continuous and strictly positive at $v = 0$. Consequently $J(t) \asymp t^{-1/4}$, with no logarithmic factor. Thus an inactive atom coinciding with a true mean contributes $1/4$, while an inactive atom away from all true means contributes $1/2$.

Product integration of (6), together with the coordinate-domain sandwich and the bounded positive density, proves

$$\int_U e^{-t\mathcal{K}_q(\theta)}\varphi(\theta)\, d\theta \asymp t^{-\lambda_{\theta^0}}, \qquad \lambda_{\theta^0} = \frac{2r-1}{2} + \frac{e}{2} + \frac{K-e-r}{4} = \frac{K+3r-2+e}{4}.$$

This determines the integrability threshold directly. Indeed, for $s > 0$, Tonelli's theorem and the Gamma integral give

$$\int_U \mathcal{K}_q(\theta)^{-s}\varphi(\theta)\,d\theta = \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\left(\int_U e^{-t\mathcal{K}_q(\theta)}\varphi(\theta)\,d\theta\right)dt.$$

The part near $t = 0$ is finite, and the part at infinity is finite exactly when $s < \lambda_{\theta^0}$. In particular the endpoint diverges. This argument does not require meromorphic continuation of the KL zeta integral.

5. Globalization. The KL divergence is continuous on Θ_I. For example, all model densities have a common Gaussian lower envelope when all means lie in I, so continuity follows by dominated convergence in the defining integral. Hence $\mathcal{Z}_q$ is compact. A finite collection of the preceding neighborhoods covers $\mathcal{Z}_q$, while KL has a strictly positive minimum on the compact complement of their union. Summing the local upper bounds and the exponentially small complementary contribution shows that the global Laplace integral is at most a constant times $t^{-(K+3r-2)/4}$.

There is a target representation with $e = 0$: allocate the K atoms among the r true means, at least one atom to each mean, and split each positive target mass into positive masses within its cluster. Its local integral provides the matching lower bound. The preceding Gamma-integral argument therefore proves the stated global coefficient and completes the proof. □

Use along a recursive trajectory. For fixed K, the preceding result gives an exact $3/4$ decrease whenever the number of distinct positive-weight components decreases by one, and a decrease of $3d/4$ when it decreases by d. This is a coefficient indexed by the target distribution, minimized over its representations in the K-component model. It need not equal the local coefficient at the current fitted representation; the local formula above specifies the difference. The interval I may be chosen separately for each target, with its true means in the interior, because the resulting minimum depends only on K and r. No single deterministic compact interval containing an entire infinite random trajectory is required for this target-indexed statement. If instead the coefficient is defined through a single global integral on $\Delta^{K-1} \times \mathbb{R}^K$, additional control of the prior tails is needed to justify globalizing the local calculation at infinity; that noncompact claim is not made here.

E Additional theory details

E.1 VaDE latent-prior Fisher bound

We prove the latent-prior Fisher scaling used in Section 2.3. For VaDE, write

$$p_\Theta(x) = \sum_{c=1}^K \pi_c q_c(x), \qquad q_c(x) = \int p_\psi(x \mid z)\mathcal{N}(z \mid \mu_c, \Sigma_c)\,dz,$$

and define

$$\rho_c(x) = \frac{\pi_c q_c(x)}{p_\Theta(x)}.$$

For a latent-prior parameter block $a_c \in \{\mu_c, \Sigma_c\}$,

$$\nabla_{a_c} \log p_\Theta(x) = \rho_c(x)\nabla_{a_c} \log q_c(x).$$

Therefore

$$I_{a_c a_c}(\Theta) = \mathbb{E}_{x\sim p_\Theta}\left[\rho_c(x)^2 \nabla_{a_c} \log q_c(x) \nabla_{a_c} \log q_c(x)^\top\right].$$

Taking traces and using $0 \le \rho_c(x) \le 1$,

$$\operatorname{tr} I_{a_c a_c}(\Theta) \le \mathbb{E}_{x\sim p_\Theta}\left[\rho_c(x)\|\nabla_{a_c} \log q_c(x)\|^2\right].$$

Since $\rho_c(x)p_\Theta(x) = \pi_c q_c(x)$, if

$$\mathbb{E}_{x\sim q_c}\left[\|\nabla_{a_c} \log q_c(x)\|^2\right] \le C_c,$$

then

$$\operatorname{tr} I_{a_c a_c}(\Theta) \le \pi_c C_c.$$

If C_c is uniformly bounded along the trajectory, this proves the upper bound $\operatorname{tr} I_{a_c a_c} = O(\pi_c)$; it is not a matching lower bound or a proportionality law.

If $E = \{c : \pi_c = 0\}$, then $p_\Theta(x)$ no longer depends on the corresponding (μ_c, Σ_c) blocks. Hence the associated score coordinates vanish identically, and

$$\dim \ker I(\Theta) \geq |E| \left(d + \frac{d(d+1)}{2} \right).$$

E.2 Proof of Proposition 1

Let $m_\alpha = \lceil \alpha n \rceil$ be the number of real samples injected at each generation. For any generation g and component k, let $N_{g,k}^{\text{real}}$ be the number of injected real samples assigned to component k. Then

$$N_{g,k}^{\text{real}} \sim \text{Binomial}(m_\alpha, \pi_k^*).$$

Since the total mixed count satisfies

$$N_{g,k}^{(\alpha)} \geq N_{g,k}^{\text{real}},$$

we have

$$\mathbb{P}\Big(N_{g,k}^{(\alpha)} = 0\Big) \leq \mathbb{P}\big(N_{g,k}^{\text{real}} = 0\big) = (1 - \pi_k^*)^{m_\alpha} \leq \exp(-m_\alpha \pi_k^*) \leq \exp(-m_\alpha \pi_{\min}^*).$$

A union bound over K components and G generations gives

$$\mathbb{P}\Big(\exists g < G, \exists k : N_{g,k}^{(\alpha)} = 0\Big) \leq KG \exp(-m_\alpha \pi_{\min}^*) \leq KG \exp(-\alpha n \pi_{\min}^*).$$

Thus, if

$$\alpha \geq \frac{\log(KG/\delta)}{n \pi_{\min}^*},$$

the failure probability is at most δ. On the complementary event, every component receives at least one mixed training sample at every generation, so no component enters the extinct set over the horizon. Hence $r_g = K$ and

$$\lambda_g = \lambda(K, K) = \frac{2K - 1}{2}, \qquad 0 \leq g \leq G.$$

E.3 Final-layer decoder curvature

Let the final decoder layer be $f_W(z) = W\phi(z)$, and consider

$$\mathcal{L}_g(W) = \frac{1}{2\sigma^2} \mathbb{E}_{(z,x)\sim Q_g} \|W\phi(z) - x\|_2^2.$$

Assume $\sigma^2 > 0$ and $\mathbb{E}_{Q_g}[\|\phi(z)\|^2 + \|x\|^2] < \infty$, and hold Q_g, ϕ, and σ^2 fixed when differentiating W. Writing W_{ij} for its entries, direct differentiation gives

$$\frac{\partial^2 \mathcal{L}_g}{\partial W_{ij} \, \partial W_{k\ell}} = \sigma^{-2} \mathbf{1}\{i = k\} \, \mathbb{E}_{Q_g}[\phi_j \phi_\ell].$$

With column-major vectorization, this is

$$\nabla^2_{\text{vec}(W)} \mathcal{L}_g(W) = \frac{1}{\sigma^2} \mathbb{E}_{Q_g}[\phi(z)\phi(z)^\top] \otimes I_{\text{out}}.$$

Therefore

$$\text{rank}\left(\nabla^2_{\text{vec}(W)} \mathcal{L}_g(W) \right) = \text{rank}\left(\mathbb{E}_{Q_g}[\phi(z)\phi(z)^\top] \right) \cdot \dim(I_{\text{out}}).$$

This identity holds with Q_g and the feature map fixed when differentiating W, using column-major vectorization. Rank loss of the uncentered feature second moment forces rank loss of this Hessian block; a decline in entropy effective rank alone does not establish loss of exact rank.

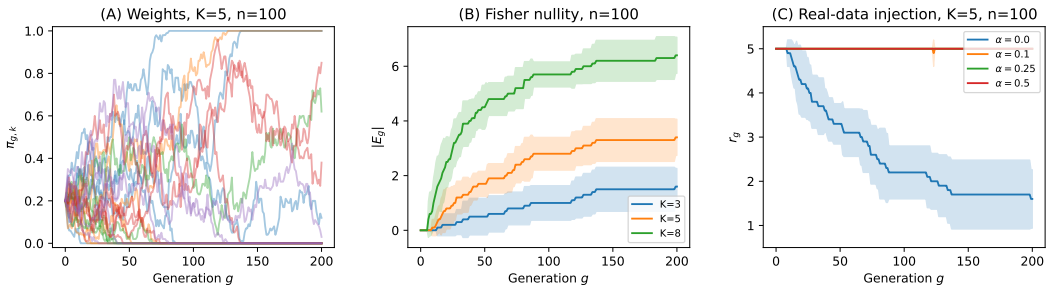


Figure 6: **Labeled univariate Gaussian-mixture recursion.** (A) Mixture-weight trajectories for $K = 5$, $n = 100$: components drift to extinction and remain absorbed. (B) The Fisher-nullity lower bound $|E_g|$ grows monotonically toward $K - 1$ (Theorem 1). (C) Real-data injection ($\alpha > 0$) arrests collapse: the tested fractions 0.25 and 0.5 prevented extinction through $G = 200$ in these runs. All panels: mean over 10 seeds, 1σ bands.

F Tier 1 — Gaussian mixtures: extinction and singular geometry

Question: *Do the observed extinction and Fisher-nullity trajectories follow the labeled-mixture mechanism?*

We run the labeled univariate Gaussian mixture recursion for $K \in \{3, 5, 8\}$, sample sizes $n \in \{50, 100, 200, 500, 1000\}$, and 10 seeds. Trajectory panels use $G = 200$ generations. The initial mixture has uniform weights $\pi_k^* = 1/K$ and separated means $\eta_k^* = 2k$.

Figure 6 records the mixture mechanism. Panel A shows absorbing multinomial dynamics: components drift, hit zero, and remain extinct. The RLCT coefficient is $(K + 3r - 2)/4$, with an exact $3/4$ decrease per extinct component (Appendix D). Its singleton values for $K = 3, 5, 8$ are 1, 1.5, and 2.25. By generation $G = 200$, the fraction of seeds reaching $r_g = 1$ is 0.70, 0.50, and 0.50 for $K = 3, 5, 8$, respectively. Panel B confirms the Fisher-nullity prediction: the mean extinct-component count $|E_g| = K - r_g$ grows monotonically from 0 to approximately 1.4, 3.4, and 6.4 at $g = 200$ for $K = 3, 5, 8$, approaching the asymptotic value $K - 1$.

Panel C shows that real-data injection arrests collapse: for $K = 5, n = 100$, synthetic-only recursion decays to $r_g \approx 1.6$ by $G = 200$, $\alpha = 0.1$ substantially slows the decay, and $\alpha \in \{0.25, 0.5\}$ prevents extinction over the full horizon, keeping $r_g = 5$. Tier 1 therefore supports the extinction, Fisher-nullity, and injection mechanisms. The exact RLCT trajectory follows from the theorem.

G Experimental details and reproducibility

This section gives the experimental details for the results in Section 3. All reported curves are averages over independent random seeds. Unless otherwise stated, shaded regions and error bars denote one standard deviation across seeds.

G.1 Data and model credits

The image experiments use MNIST [LeCun et al., 1998], Fashion-MNIST [Xiao et al., 2017], and CIFAR-10 [Krizhevsky, 2009]. The language experiment uses OPT-125M [Zhang et al., 2022] and WikiText-2 [Merity et al., 2016]. CIFAR-10 was created by Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton. Original dataset distribution pages are `https://yann.lecun.com/exdb/mnist/`, `https://github.com/zalandoresearch/fashion-mnist`, and `https://www.cs.toronto.edu/~kriz/cifar.html`.

The Fashion-MNIST repository carries the MIT license (Zalando SE, 2017). Official OPT model weights are governed by Meta's custom OPT-175B License Agreement, which permits non-commercial research subject to its restrictions and redistribution conditions; this model-weight license is distinct from the accompanying software license. WikiText-2 is distributed under Creative

Commons Attribution-ShareAlike terms. The authoritative OPT terms and WikiText dataset card are available at `https://github.com/facebookresearch/metaseq/blob/main/projects/OPT/MODEL_LICENSE.md` and `https://huggingface.co/datasets/Salesforce/wikitext`.

G.2 Datasets and recursive protocol

For the Gaussian-mixture tier, data are generated synthetically from the specified labeled Gaussian mixture. VaDE and the standard VAE are trained on MNIST, and the compact DDPM is trained on FashionMNIST. In these three settings, images are resized or kept at 28×28, normalized to $[0, 1]$, and flattened for MLP-based VaDE/VAE models; the compact DDPM uses the image format directly. The larger diffusion models use CIFAR-10 (Appendix I).

Across tiers, recursive training follows

$$\mathcal{D}_{g+1} \sim p_{\theta_g}^{\otimes n}, \qquad \theta_{g+1} = \mathrm{Train}(\mathcal{D}_{g+1}, \theta_g).$$

The α-mixed control replaces an α-fraction of each generation's synthetic samples with fresh samples from the original training distribution:

$$\mathcal{D}_{g+1}^{(\alpha)} = \mathcal{D}_{g+1}^{\mathrm{syn}} \cup \mathcal{D}_{g+1}^{\mathrm{real}}, \qquad |\mathcal{D}_{g+1}^{\mathrm{real}}| = \lceil \alpha n \rceil.$$

We use $\alpha = 0$ for synthetic-only recursion. VaDE and VAE experiments use $\alpha \in \{0, 0.1, 0.25, 0.5\}$, while the DDPM experiment uses $\alpha \in \{0, 0.1, 0.25\}$.

G.3 Tier 1: Gaussian mixtures

For the labeled Gaussian-mixture tier, we use a K-component univariate Gaussian mixture with known unit variance. The initial mixture has uniform weights

$$\pi_k^* = 1/K$$

and separated means

$$\eta_k^* = 2k, \qquad k = 1, \ldots, K.$$

The labeled recursion uses the closed-form MLE update described in Section 2.1. We run

$$K \in \{3, 5, 8\}, \qquad n \in \{50, 100, 200, 500, 1000\},$$

with 10 random seeds. Trajectory plots use $G = 200$ recursive generations. The theoretical RLCT trajectory is defined by

$$\lambda_g = \lambda(K, r_g),$$

using the formula in Section 2.2.

G.4 Tier 2: VaDE

For VaDE, we use $K = 10$ latent mixture components and latent dimension $d = 10$. The encoder and decoder are multilayer perceptrons. The encoder outputs the variational Gaussian parameters $(\mu_\phi(x), \log \sigma_\phi^2(x))$, and the decoder maps $z \in \mathbb{R}^{10}$ back to the image space. The latent prior is

$$p(z, c) = \pi_c \mathcal{N}(z \mid \mu_c, \Sigma_c).$$

We train on MNIST with

$$n = 5000, \qquad G = 50, \qquad 10 \text{ seeds}, \qquad \alpha \in \{0, 0.1, 0.25, 0.5\}.$$

Each generation is trained by ELBO maximization initialized from the previous generation's weights unless otherwise specified. Components with

$$\pi_{g,k} < 10^{-3}$$

are treated as near-extinct, giving

$$E_g^\varepsilon = \{k : \pi_{g,k} < 10^{-3}\}, \qquad r_g^\varepsilon = K - |E_g^\varepsilon|.$$

For each component k, we compute the normalized latent-prior Fisher curvature

$$\kappa_{g,k} := \frac{1}{d + d(d+1)/2} \operatorname{tr} I_{(\mu_{g,k}, \Sigma_{g,k}), (\mu_{g,k}, \Sigma_{g,k})}(\Theta_g).$$

The candidate low-curvature subspace dimension is

$$|E_g^\varepsilon| \left(d + \frac{d(d+1)}{2} \right),$$

which equals $65|E_g^\varepsilon|$ for $d = 10$.

G.5 Tier 3: standard VAE

The standard VAE uses the same MLP encoder/decoder architecture and latent dimension $d = 10$ as the VaDE tier, but replaces the Gaussian-mixture latent prior with a single standard Gaussian prior. We train on MNIST with

$$n = 5000, \qquad G = 50, \qquad 10 \text{ seeds}, \qquad \alpha \in \{0, 0.1, 0.25, 0.5\}.$$

Each recursive generation uses 50 epochs of continuation training. Because the standard VAE has no explicit mixture weights, we evaluate collapse using class-coverage entropy, empirical LLC, Hessian/Gauss–Newton effective ranks, and decoder feature-covariance effective rank rather than latent-component extinction.

G.6 Tier 4: DDPM

The DDPM tier uses a compact U-Net denoising model trained on FashionMNIST. We use

$$G = 20, \qquad 5 \text{ seeds}, \qquad \alpha \in \{0, 0.1, 0.25\}.$$

Each generation trains the denoising model on the generated image set, with the α-mixed control replacing an α-fraction of generated samples with fresh FashionMNIST samples. Distributional collapse is measured by class-coverage entropy and FID. Geometry is measured by the effective rank of U-Net feature covariances and a blockwise empirical Gauss–Newton effective rank computed on the denoising loss.

G.7 Diagnostics

For image models, class-coverage entropy is computed as

$$H_{\mathrm{cov}}(g) = -\sum_{c=1}^{10} q_{g,c} \log q_{g,c},$$

where $q_{g,c}$ is the fraction of generated samples classified as class c by a fixed classifier trained on the corresponding real dataset. For MNIST experiments, the classifier is trained on real MNIST; for FashionMNIST experiments, it is trained on real FashionMNIST.

For a positive semidefinite matrix with eigenvalues $\{\lambda_i\}_{i=1}^m$, we compute effective rank using the entropy definition

$$r_{\mathrm{eff}} = \exp\left(-\sum_{i=1}^{m} \tilde{\lambda}_i \log \tilde{\lambda}_i\right), \qquad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j}.$$

The metric utility clips negative eigenvalues to zero before computing entropy effective rank and assigns rank zero to an all-zero spectrum. Its normalized version divides by the number of supplied eigenvalues.

Hessian and Gauss–Newton ranks are computed on minibatches from the generation-g training or evaluation set. The feature utility centers the latent features $\phi(z)$ entering the final decoder layer:

$$\widehat{\Sigma}_{\phi,g} = \frac{1}{m-1} \sum_{i=1}^{m} (\phi(z_i) - \bar{\phi})(\phi(z_i) - \bar{\phi})^\top, \qquad \bar{\phi} = \frac{1}{m} \sum_{i=1}^{m} \phi(z_i).$$

This centered covariance differs from the uncentered second moment in the exact decoder-curvature identity by a mean outer product (and the finite-sample normalization). For the DDPM, U-Net feature rank is computed analogously from intermediate U-Net activations, and the blockwise Gauss–Newton rank is computed from selected parameter blocks of the denoising objective.

G.8 Empirical LLC estimation

Empirical local learning coefficient estimates are computed using the SGLD-based estimator of Furman and Lau [Furman and Lau, 2024]. For each checkpoint, we run SGLD chains initialized near the trained parameter θ_g, sample parameter perturbations from the local posterior, and estimate the local learning coefficient from the temperature-scaled increase in empirical loss. Robustness checks repeat the LLC computation with doubled chain length and step-size perturbations, as shown in Figure 5D.

G.9 Early-warning prediction

For VaDE, the early-warning target at generation g is

$$\mathcal{C}_{g,\Delta} = \mathbf{1}\{r^{\varepsilon}_{g+\Delta} < r^{\varepsilon}_{g}\}, \qquad \Delta = 10.$$

We train logistic predictors using three feature groups: distribution-only features, geometry-only features, and their union. Distribution-only features include active component count, class-coverage entropy, and held-out negative log-likelihood. Geometry-only features include componentwise latent-prior Fisher curvature, Fisher near-nullity, empirical LLC, Hessian effective rank, Gauss–Newton effective rank, and decoder feature-covariance rank. Prediction is evaluated using leave-one-seed-out splits, so checkpoints from the same recursive trajectory never appear in both the training and test folds. We report mean AUROC and one standard deviation across held-out-seed folds.

G.10 Statistical reporting

Reported means and uncertainty bands summarize the evaluation runs described for each experiment. Unless otherwise stated, curves and tables report mean $\pm$ one standard deviation. The Gaussian-mixture, VaDE, and VAE experiments use 10 seeds. The DDPM experiment uses 5 seeds because each recursive diffusion run is more computationally expensive. No normality assumption is used to interpret the bands as confidence intervals.

G.11 Implementation

The supplementary archive contains runnable labeled-GMM recursion, metric utilities, a leave-one-seed-out VaDE prediction script, CSV schemas, and plotting scripts. The predictor standardizes features within each training fold and fits class-balanced logistic regression with a maximum of 2000 iterations.

G.12 Compute resources

The Gaussian-mixture experiments are lightweight and can be run on CPU. The VaDE and VAE experiments can be run on a single GPU or CPU, although GPU training is faster. The DDPM experiments are run on GPU. Approximate runtimes depend on hardware and implementation details; in our runs, the Gaussian-mixture tier required minutes on CPU, VaDE/VAE recursive runs required hours on a single GPU, and the DDPM recursive runs required substantially longer because a denoising model is trained at each generation.

H Additional neural bridge results

Throughout, $p_\Theta(x) = \sum_{k=1}^{K} \pi_k q_k(x)$, $\rho_k(x) = \pi_k q_k(x)/p_\Theta(x)$, and all displayed expectations are finite and the stated derivatives exist. The q_k are densities relative to a common dominating measure; responsibilities may be defined arbitrarily on the p_Θ-null set where $p_\Theta = 0$. Component-specific parameters occupy disjoint coordinate blocks; differentiation under the defining integrals is justified by an integrable dominating function. Eigenvalues $\lambda_j^{\uparrow}$ are ordered increasingly.

H.1 Softmax coordinates and ideal responsibility recursion

Proposition 2 (Softmax Fisher information)**.** *Hold the component marginals fixed and write* $\pi = \mathrm{softmax}(\beta)$. *Then*

$$\nabla_\beta \log p_\Theta(X) = \rho(X) - \pi, \qquad I_\beta = \mathrm{Cov}_{p_\Theta}(\rho(X)) \preceq \mathrm{Diag}(\pi) - \pi\pi^\top. \tag{7}$$

In particular, $I_{\beta_k\beta_k} \le \pi_k(1-\pi_k) \le \pi_k$, *and* $I_\beta \mathbf{1} = 0$. *The constant-shift null direction is present at every interior point and can be removed by restricting to* $\mathbf{1}^\perp$.

Proof. Differentiating π_j gives $\partial\pi_j/\partial\beta_k = \pi_j(\mathbf{1}\{j=k\} - \pi_k)$, which yields the score identity. If Z is the one-hot component indicator, then $\rho(X) = \mathbb{E}[Z \mid X]$ and $\mathbb{E}\rho(X) = \pi$. The law of total covariance gives

$$\mathrm{Cov}(Z) = \mathrm{Cov}(\rho(X)) + \mathbb{E}[\mathrm{Cov}(Z \mid X)],$$

proving the matrix bound. The diagonal bound follows immediately, and the score sums to zero. $\square$

Finite logits represent only the simplex interior. The limit $\pi_k \to 0$ requires $\beta_k - \log \sum_{j \neq k} \exp(\beta_j) \to -\infty$; it does not require an absolute logit to diverge in a particular gauge. For an orthonormal basis matrix Q of $\mathbf{1}^\perp$, the Fisher matrix after removing the gauge is $Q^\top I_\beta Q$. More generally, under $a = h(b)$ with nonsingular Jacobian J, Fisher information transforms as $I_b = J^\top I_a J$, preserving rank and nullity. Uniformly bounded J preserves $O(\pi_k)$ small-eigenvalue upper bounds; two-sided norm comparison also requires a uniformly bounded inverse. Local RLCTs are invariant under nonsingular analytic coordinate changes with the prior transformed accordingly, or with positive comparable local prior densities. These facts do not assert invariance of individual eigenvalues or finite-sample localized LLC estimates, and a softmax chart does not extend nonsingularly to the simplex boundary.

Proposition 3 (Unlabeled responsibility update)**.** *Let n be fixed. Conditional on $\mathcal{F}_g$, let the current weights and component marginals be fixed and draw $X_1, \ldots, X_n$ independently from p_{Θ_g}. Define*

$$\widetilde{\pi}_{g+1} = \frac{1}{n} \sum_{i=1}^{n} \rho_g(X_i), \qquad D(\pi) = 1 - \|\pi\|_2^2.$$

Write $I_{\beta,g} = \mathrm{Cov}_{p_{\Theta_g}}(\rho_g(X))$, using this extension of (7) *also for boundary weights. Then*

$$\mathbb{E}[D(\widetilde{\pi}_{g+1}) \mid \mathcal{F}_g] = D(\pi_g) - \frac{1}{n} \operatorname{tr} I_{\beta,g}. \tag{8}$$

Under exact iteration $\pi_{g+1} = \widetilde{\pi}_{g+1}$, the weights form a bounded martingale, $D(\pi_g)$ is a nonnegative supermartingale, and $\sum_{g \geq 0} \operatorname{tr} I_{\beta,g} < \infty$ almost surely. If, additionally, for some fixed $\Delta > 0$, every pair of indices $k \neq l$ with $\pi_{g,k}\pi_{g,l} > 0$ satisfies $\|q_{g,k} - q_{g,l}\|_{L^1} \geq \Delta$ at every generation, then $D(\pi_g) \to 0$ almost surely: the limiting weight vector is a vertex. For an approximate update $\pi_{g+1} = \widetilde{\pi}_{g+1} + e_g$, the absolute perturbation of the conditional expectation in (8) *is at most*

$$2\mathbb{E}[\|e_g\|_2 \mid \mathcal{F}_g] + \mathbb{E}[\|e_g\|_2^2 \mid \mathcal{F}_g]. \tag{9}$$

Proof. By Proposition 2, $\mathbb{E}[\widetilde{\pi}_{g+1} \mid \mathcal{F}_g] = \pi_g$ and $\mathrm{Cov}(\widetilde{\pi}_{g+1} \mid \mathcal{F}_g) = I_{\beta,g}/n$. Expanding the squared norm proves (8) and the martingale claims. Telescoping expectations and monotone convergence give

$$\mathbb{E} \sum_{g \geq 0} \operatorname{tr} I_{\beta,g} \leq n\mathbb{E}D(\pi_0) < \infty,$$

so the series is finite almost surely and its terms tend to zero.

For the separation assertion, suppose $K \geq 2$ and $D(\pi) > 0$ at a given generation. Choose $a \neq b$ with $\pi_a \pi_b \geq D(\pi)/[K(K-1)]$. Cauchy–Schwarz and the score identity imply

$$\operatorname{tr} I_\beta = \sum_{k:\pi_k > 0} \pi_k^2 \int \frac{(q_k - p)^2}{p} \geq \sum_{k:\pi_k > 0} \pi_k^2 \|q_k - p\|_{L^1}^2.$$

Writing $A = \|q_a - p\|_{L^1}$ and $B = \|q_b - p\|_{L^1}$, separation gives $A + B \geq \Delta$. Since $\pi_a, \pi_b \leq 1$,

$$\operatorname{tr} I_\beta \geq (\pi_a \pi_b)^2 (A^2 + B^2) \geq \frac{\Delta^2 D(\pi)^2}{2K^2(K-1)^2}.$$

Consequently $\operatorname{tr} I_{\beta,g} \to 0$ forces $D(\pi_g) \to 0$; convergence of the bounded weight martingales identifies a vertex limit. The case $K = 1$ is immediate. Finally,

$$D(\widetilde{\pi}_{g+1} + e_g) - D(\widetilde{\pi}_{g+1}) = -2\widetilde{\pi}_{g+1}^\top e_g - \|e_g\|_2^2,$$

and $\|\widetilde{\pi}_{g+1}\|_2 \leq 1$ proves (9). □

The separation assumption is substantive: identical component marginals give $I_\beta = 0$ and stationary interior weights.

H.2 Multiplicity of component-induced small Fisher eigenvalues

Proposition 4 (Full-Fisher near-null multiplicity)**.** *Let $I(\Theta)$ be finite and let $a_k \in \mathbb{R}^{d_k}$ be disjoint component-specific parameter blocks. For a nonempty component set E, assume*

$$s_k(x) = \nabla_{a_k} \log q_k(x), \qquad \mathbb{E}_{q_k}[s_k s_k^\top] \preceq C_k I_{d_k}, \qquad k \in E.$$

Set $d_E = \sum_{k\in E} d_k$ and $\delta_E = \max_{k\in E} \pi_k C_k$. Then

$$\lambda^{\uparrow}_{d_E}(I(\Theta)) \le \delta_E. \tag{10}$$

Thus at least d_E full-Fisher eigenvalues are no larger than δ_E. At exact extinction these directions are null. Along a trajectory the sufficient condition for their curvature to vanish is $\delta_E \to 0$; bounded component score moments make $\pi_k \to 0$ sufficient.

Proof. The component score identity gives $\nabla_{a_k} \log p = \rho_k s_k$. For a vector v supported on the indicated blocks, weighted Cauchy–Schwarz and $\sum_{k\in E} \rho_k \le 1$ yield

$$\begin{aligned} v^\top I(\Theta) v &= \mathbb{E}_p\Big(\sum_{k\in E} \rho_k v_k^\top s_k\Big)^2 \\ &\le \mathbb{E}_p \sum_{k\in E} \rho_k (v_k^\top s_k)^2 = \sum_{k\in E} \pi_k \mathbb{E}_{q_k}(v_k^\top s_k)^2 \le \delta_E \|v\|_2^2. \end{aligned}$$

The min–max principle on this d_E-dimensional subspace proves the bound. If the corresponding weights vanish, their score coordinates vanish identically. □

Lemma 4 (A sufficient Gaussian score bound)**.** *For $q_k(x) = \int p_\psi(x \mid z)\mathcal{N}(z \mid \mu_k, \Sigma_k)\, dz$, hold ψ fixed and use direct mean coordinates and an orthonormal basis for the symmetric covariance matrices. If $\Sigma_k \succeq s^2 I$ with $s > 0$, the joint mean/covariance score satisfies the condition of Proposition 4 with $C_k \le \max\{s^{-2}, (2s^4)^{-1}\}$.*

Proof. Differentiation under the integral expresses the component score as the conditional expectation of the Gaussian latent score given X. Conditional Jensen bounds its information by that of the latent Gaussian. For a mean direction h and symmetric covariance direction A, the latter directional information equals

$$h^\top \Sigma_k^{-1} h + \frac{1}{2}\operatorname{tr}(\Sigma_k^{-1} A \Sigma_k^{-1} A) \le s^{-2}\|h\|_2^2 + \frac{1}{2s^4}\|A\|_F^2.$$

The mean/covariance cross term is zero by the vanishing Gaussian third moments. The assumed coordinate basis identifies $\|A\|_F$ with its coordinate norm, proving the claim. □

Other fixed covariance coordinate conventions change only a constant. For full covariance and latent dimension d, each component supplies $d + d(d+1)/2$ directions, equal to 65 when $d = 10$. A small-weight threshold selects a candidate subspace; a numerical eigenvalue tolerance τ is guaranteed only when $\delta_E \le \tau$. These are upper bounds, not assertions that Fisher curvature is proportional to component weight.

H.3 From output Jacobians to blockwise curvature

Proposition 5 (Jacobian-Gram control of generalized Gauss–Newton curvature)**.** *For a specified parameter block W, let J_W be the Jacobian of the model output u entering a twice-differentiable loss, and let its output Hessian B satisfy $0 \preceq B \preceq LI$ almost surely under a fixed data distribution. Then*

$$G_W := \mathbb{E}[J_W^\top B J_W] \preceq L A_W, \qquad A_W := \mathbb{E}[J_W^\top J_W], \tag{11}$$

and $\lambda_j^{\uparrow}(G_W) \le L\lambda_j^{\uparrow}(A_W)$ for every j. For a direct dense output $u = W\phi$, with ϕ fixed when differentiating W, $A_W = M_\phi \otimes I_{\text{out}}$, where $M_\phi = \mathbb{E}[\phi\phi^\top]$. Thus m eigenvalues of M_ϕ no larger than ε imply at least md_{out} eigenvalues of G_W no larger than $L\varepsilon$.

Proof. For every vector v, the output-Hessian bound gives $v^\top G_W v \le L\mathbb{E}\|J_W v\|_2^2 = Lv^\top A_W v$. The min–max principle proves the eigenvalue comparison. With column-major vectorization of W, the dense output has $J_W = \phi^\top \otimes I_{\text{out}}$. Its Gram is $\phi\phi^\top \otimes I_{\text{out}}$, whose feature eigenvalues repeat d_{out} times. □

For independent Bernoulli likelihoods in output logits, $B = \text{Diag}(\text{sigmoid}(u_i)(1-\text{sigmoid}(u_i))) \preceq I/4$. For the half-squared output error, $B = I$. Loss averaging or weighting rescales these constants. For a final linear convolution, A_W is the corresponding aggregate patch Gram with shared weights. For an internal convolutional block, J_W must include the downstream network derivatives.

The exact squared-loss final-layer Hessian identity in Appendix E.3 is the special case $B = I/\sigma^2$ with frozen features and sampling distribution. In general the full neural Hessian includes additional second derivatives of the model output and need not be positive semidefinite. Also, M_ϕ is an uncentered second moment: $M_\phi = \text{Cov}(\phi) + \mathbb{E}[\phi]\mathbb{E}[\phi]^\top$. For a joint weight-and-bias block, the same statements use augmented features $(\phi, 1)$. Finally, decreasing entropy effective rank describes concentration of normalized eigenvalues; it does not by itself establish small absolute eigenvalues or an RLCT decrease.

H.4 A conditional full-model RLCT comparison

Proposition 6 (Nested loss germs and transverse constraint deletion)**.** *Let K_{old} and K_{new} be nonnegative analytic loss germs vanishing at the origin in a common local coordinate space. Use local prior densities bounded above and below by positive constants. If $K_{\text{new}} \le CK_{\text{old}}$ on a sufficiently small neighborhood, then $\lambda_{\text{new}} \le \lambda_{\text{old}}$. In particular, this comparison holds when each new analytic residual is an analytic linear combination of the old residuals and the losses are comparable to the respective sums of squared residuals. If, in local analytic coordinates (u, v) with $u \in \mathbb{R}^d$, the stronger transverse product condition holds,*

$$K_{\text{old}}(u, v) \asymp \|u\|_2^2 + K_{\text{surv}}(v), \qquad K_{\text{new}}(u, v) \asymp K_{\text{surv}}(v), \tag{12}$$

with K_{surv} a nonnegative analytic germ, then $\lambda_{\text{old}} - \lambda_{\text{new}} = d/2$. Here $\asymp$ denotes two-sided comparison by positive local constants; the RLCT of an identically zero germ is taken to be zero.

Proof. If $K_{\text{old}} \equiv 0$, the comparison forces $K_{\text{new}} \equiv 0$; if only $K_{\text{new}} \equiv 0$, monotonicity is immediate. Otherwise, a nonzero germ's local RLCT is the integrability threshold

$$\lambda(K) = \sup\left\{ s \ge 0 : \int_U K(w)^{-s}\varphi(w)\, dw < \infty \right\}$$

on a sufficiently small neighborhood U. The inequality $K_{\text{new}} \le CK_{\text{old}}$ implies that integrability of K_{new}^{-s} entails integrability of K_{old}^{-s}, proving monotonicity. Comparable positive priors do not change this threshold. If the new residual vector is $A(w)$ times the old one, the analytic matrix A is bounded after shrinking U, giving the required squared-loss comparison.

For (12), work on a small product neighborhood. Up to constant factors and a constant rescaling of t, the old local Laplace integral is

$$\int e^{-t(\|u\|_2^2 + K_{\text{surv}}(v))}\, du\, dv = \left(\int e^{-t\|u\|_2^2}\, du\right)\left(\int e^{-tK_{\text{surv}}(v)}\, dv\right).$$

The first factor has order $t^{-d/2}$; the second has leading power $t^{-\lambda_{\text{surv}}}$, possibly with logarithmic factors. The new loss has the latter power alone because the u integral is a constant volume. Hence $\lambda_{\text{old}} = d/2 + \lambda_{\text{surv}}$ and $\lambda_{\text{new}} = \lambda_{\text{surv}}$. If $K_{\text{surv}} \equiv 0$, its Laplace factor is a constant volume, giving the same conclusion with $\lambda_{\text{surv}} = 0$. □

This proposition is conditional on comparable germs in common coordinates and, for the exact drop, on (12); deleting merely linearly independent residuals is not sufficient to assert the product condition. Shared neural retraining can change the surviving representation and loss germ, so nesting is not automatic. Nonsmooth architectures also require an appropriate analytic stratum or a separate extension of the argument. For example, on a common two-dimensional neighborhood with a positive prior, $K_{\text{old}}(u, v) = u^2$ has RLCT $1/2$, whereas $K_{\text{new}}(u, v) = u^2 + v^2$ has RLCT 1; arbitrary changes to the loss can increase the RLCT.

I Additional controls, intervention, and scale experiments

This section gives the numerical results and evaluation protocols for the fixed-real controls, geometry-triggered intervention, dose transfer, and larger-model experiments. Unless otherwise stated, values are means; bootstrap intervals are identified separately. The compact diffusion model is the 1.8M-parameter FashionMNIST DDPM. The larger 12M-, 40M-, and 96M-parameter diffusion models use CIFAR-10.

I.1 Diagnostic glossary

Coverage and support. For a fixed classifier trained only on real data, $q_{g,c}$ is the generated prevalence of class c. Class-coverage entropy is $-\sum_c q_{g,c} \log q_{g,c}$, using natural logarithms. A class is active when its generated prevalence exceeds 1%. VaDE latent components instead use the near-extinction threshold $\pi_{g,k} < 10^{-3}$. These quantities measure different notions of support: generated class coverage and latent mixture support need not coincide.

Effective rank and near-nullity. For a positive semidefinite matrix with nonzero eigenvalue sum, effective rank is $\exp(-\sum_i \widetilde{\lambda}_i \log \widetilde{\lambda}_i)$, where $\widetilde{\lambda}_i = \lambda_i / \sum_j \lambda_j$. A smaller effective rank indicates greater spectral concentration. The implementation normalizes by the number of eigenvalues, as described in Appendix G. Fisher near-nullity counts parameter directions with small Fisher curvature; it is distinct from entropy-based effective rank. The theoretical spectral bounds concern small eigenvalues, whereas the reported effective-rank changes are empirical measurements. Blockwise generalized Gauss–Newton (GGN) rank measures the corresponding curvature spectrum in a selected neural parameter block.

Local learning coefficient. The empirical LLC is a local, finite-sample complexity estimate obtained with localized tempered SGLD. Full-model LLC perturbs the full parameter vector; block-local LLC holds the remaining parameters fixed.

I.2 Consolidated four-tier diagnostic endpoints

Table 4 collects the principal distributional and geometric endpoints for the four original tiers. Each tier uses its own reported horizon. The larger CIFAR-10 DDPMs and the language-model experiment are reported separately in Tables 12, 15, and 16.

The tables use initialization, synthetic-only final, and fixed $\alpha = 0.25$ final conditions. GMM uses $K = 5$, $n = 100$, $G = 200$; VaDE/VAE use $G = 50$; the compact FashionMNIST DDPM uses $G = 20$. N/A denotes a diagnostic not applicable to the tier as evaluated.

At $\alpha = 0.5$, VaDE retains 9.7 ± 0.3 active components at generation 50. For the standard VAE at generation 50 and $\alpha \geq 0.25$, entropy remains above 2.05, LLC above 98.0, and feature rank above 7.4. The final normalized collapse indices for synthetic-only GMM, VaDE, and VAE are 0.82 ± 0.07, 0.74 ± 0.08, and 0.69 ± 0.07, respectively, reaching means of approximately 0.2 or below at $\alpha \geq 0.25$ (Figure 5A).

I.3 Matched fixed-real-data controls

For each seed, synthetic recursion and the fixed-real control begin from the same generation-0 checkpoint. Synthetic recursion draws a new model-generated dataset at every stage; the control reuses one real dataset of matched size:

$$\mathcal{D}_{g+1}^{\mathrm{syn}} \sim p_{\theta_g}^{\otimes n}, \qquad \mathcal{D}^{\mathrm{fixed}} \sim (p^*)^{\otimes n} \text{ once}, \qquad \mathcal{D}_{g+1}^{\mathrm{control}} = \mathcal{D}^{\mathrm{fixed}}.$$

Both conditions continue training from θ_g. Architecture, initialization, optimizer, learning rate, minibatch size, updates per stage, checkpoint schedule, evaluation split, and seeds are matched, so only the next-stage data source differs. VaDE and VAE use MNIST and 10 seeds; the compact DDPM uses FashionMNIST and 5 seeds. Unlike the $\alpha = 1$ condition, which draws fresh real observations at each stage, this control repeatedly uses the same observations. For the labeled GMM, repeated labeled MLE on a fixed dataset returns the same estimate after the initial fit, making all subsequent support and RLCT values stationary.

Table 4: **Four-tier support, complexity, and representation summary.** Support means active-component count for GMM/VaDE and class-coverage entropy for VAE/DDPM. Feature rank is normalized for DDPM only.

Tier	Condition	Support/entropy	LLC	Feature rank
GMM	Initial	5	NR	N/A
	Synthetic	≈ 1.6	NR	N/A
	$\alpha = 0.25$	5	NR	N/A
VaDE	Initial	10.0 ± 0.0	128.4 ± 6.7	8.9 ± 0.4
	Synthetic	3.6 ± 0.8	80.1 ± 8.9	3.9 ± 0.7
	$\alpha = 0.25$	8.9 ± 0.6	119.6 ± 6.2	NR
VAE	Initial	2.30 ± 0.0062	116.2 ± 5.8	8.6 ± 0.5
	Synthetic	1.21 ± 0.18	69.4 ± 7.6	3.7 ± 0.6
	$\alpha = 0.25$	> 2.05	108.9 ± 5.7	7.6 ± 0.5
DDPM	Initial	2.29 ± 0.02	NR	0.94 ± 0.02
(1.8M)	Synthetic	1.37 ± 0.05	NR	0.51 ± 0.02
	$\alpha = 0.25$	2.28 ± 0.05	NR	0.96 ± 0.02

Table 5: **Four-tier curvature and sample-quality summary.** DDPM GGN rank is normalized. Fisher dimension is the extinct-component nullity lower bound for GMM; for VaDE it is the threshold-selected $65|E_g^{\varepsilon}|$ candidate subspace dimension.

Tier	Condition	GGN rank	Fisher dimension	FID
GMM	Initial	N/A	0	N/A
	Synthetic	N/A	≈ 3.4	N/A
	$\alpha = 0.25$	N/A	0	N/A
VaDE	Initial	NR	NR	N/A
	Synthetic	NR	416 ± 52	N/A
	$\alpha = 0.25$	NR	NR	N/A
VAE	Initial	NR	N/A	N/A
	Synthetic	NR	N/A	N/A
	$\alpha = 0.25$	NR	N/A	N/A
DDPM	Initial	0.92 ± 0.03	N/A	33.2 ± 1.1
(1.8M)	Synthetic	0.53 ± 0.04	N/A	90.0 ± 2.3
	$\alpha = 0.25$	0.89 ± 0.03	N/A	29.9 ± 1.0

Table 6: Matched fixed-real controls. The DDPM in this table is the compact FashionMNIST model. Feature and GGN ranks for this model are normalized.

Model	Diagnostic	Initial	Synthetic final	Fixed-real final
VaDE	Active components	10.0 ± 0.0	3.6 ± 0.8	9.8 ± 0.3
	LLC	128.4 ± 6.7	80.1 ± 8.9	124.9 ± 5.9
	Decoder feature rank	8.9 ± 0.4	3.9 ± 0.7	8.6 ± 0.5
VAE	Coverage entropy	2.30 ± 0.0062	1.21 ± 0.18	2.26 ± 0.04
	LLC	116.2 ± 5.8	69.4 ± 7.6	112.8 ± 6.1
	Decoder feature rank	8.6 ± 0.5	3.7 ± 0.6	8.1 ± 0.4
DDPM	Coverage entropy	2.29 ± 0.02	1.37 ± 0.05	2.24 ± 0.04
	FID	33.2 ± 1.1	90.0 ± 2.3	34.1 ± 1.5
	U-Net feature rank	0.94 ± 0.02	0.51 ± 0.02	0.93 ± 0.02
	Blockwise GGN rank	0.92 ± 0.03	0.53 ± 0.04	0.88 ± 0.03

Paired-seed comparisons give a fixed-real minus synthetic difference of 44.8 LLC units for VaDE (95% bootstrap CI $[38.7, 50.9]$), and 1.05 entropy units for the VAE ($[0.92, 1.18]$). Compact-DDPM FID is 55.9 lower under fixed-real training ($[52.1, 59.8]$). Thus the additional degeneration is associated with recursively replacing observations by model-generated samples, rather than with the cumulative optimization budget alone.

Held-out performance remains close to initialization under the fixed-real control (Table 7), providing an additional check on generalization while the same training observations are reused.

Table 7: Held-out performance for matched controls; lower is better for each metric.

Model	Metric	Initial	Synthetic final	Fixed-real final
VaDE	NLL	86.2 ± 1.4	121.8 ± 4.9	87.1 ± 1.6
VAE	NLL	101.4 ± 2.2	149.6 ± 6.1	103.0 ± 2.5
Compact DDPM	FID	33.2 ± 1.1	90.0 ± 2.3	34.1 ± 1.5

I.4 Softmax-coordinate checks and smooth trajectories

The VaDE experiments use softmax mixture weights. Across $5{,}100$ component–generation–seed checkpoints, the measured logit-coordinate Fisher diagonals satisfy $I_{\beta_k\beta_k} \le \pi_k(1-\pi_k)$. The ratio $I_{\beta_k\beta_k}/[\pi_k(1-\pi_k)]$ has median 0.43, 95th percentile 0.79, and maximum 0.96. Removing the permanent constant-shift gauge, centered-logit coordinates and an orthonormal simplex-tangent basis give identical rank trajectories; normalized-curvature trajectories have mean Spearman correlation 0.98 ± 0.01 across seeds. Component-specific latent-prior curvature scales with weight with log–log slope 0.94 and $R^2 = 0.85$.

The closed-simplex extinction statement and the finite-softmax measurements describe different stages of the same limiting mechanism. Finite logits represent positive weights, with extinction approached as the relevant logit differences diverge. Consequently, VaDE curvature, effective ranks, and LLC estimates can change continuously before a near-extinction threshold is crossed. Active component count is integer-valued for each seed, but averaging threshold crossings at different generations can smooth its trajectory. Exact GMM extinction instead produces discrete changes in support and in the closed-form RLCT. The standard VAE and DDPM have no mixture-weight extinction mechanism; their geometry diagnostics probe representation and curvature degeneration.

I.5 Early warning beyond explicit mixture components

For the standard VAE and compact DDPM, fixed classifiers trained only on real MNIST and Fashion-MNIST, respectively, determine generated class prevalences. Predictors use development trajectories and are evaluated on held-out seeds. With c_g denoting active-class count, the reported prediction target is a decrease in count within the next Δ generations, with $\Delta = 10$ for the VAE and $\Delta = 5$ for the DDPM. This event is

$$\mathcal{C}^{\mathrm{window}}_{g,\Delta} = \mathbf{1}\{\min_{1\le h\le \Delta} c_{g+h} < c_g\}.$$

The original VaDE evaluation instead uses the endpoint component-count target $\mathbf{1}\{r_{g+\Delta} < r_g\}$ and the leave-one-seed-out protocol in Appendix G. These events can differ when a count falls and then recovers. Geometry contributes predictive information beyond the distributional diagnostics in all three models.

I.6 Geometry-triggered real-data intervention

After generation g, a predictor frozen on validation data estimates collapse risk over the next Δ generations. If risk exceeds the alarm threshold and budget remains, generation $g+1$ uses real-data fraction $\alpha_{\mathrm{on}} = 0.25$; otherwise it uses zero. The forecast horizon is $\Delta = 10$, except for DDPM, where $\Delta = 5$. At matched average real-data budget $B = 0.10$, we compare geometry-triggered, distribution-triggered, random-triggered, and fixed-$\alpha = 0.10$ policies, alongside synthetic-only trajectories. Fixed $\alpha = 0.25$ is a higher-budget reference. The geometric policy uses 60% fewer real examples than this higher-budget reference.

Table 8: Held-out early-warning AUROC. The compact DDPM uses FashionMNIST; the three larger DDPMs use CIFAR-10, $G = 20$, $\Delta = 5$, and five held-out seeds per size.

Model	Distribution only	Geometry only	Combined
VaDE	0.72 ± 0.04	0.85 ± 0.03	0.95 ± 0.01
VAE	0.73 ± 0.04	0.84 ± 0.03	0.93 ± 0.02
Compact DDPM (1.8M)	0.70 ± 0.05	0.81 ± 0.04	0.90 ± 0.03
CIFAR-10 DDPM (12M)	0.71 ± 0.04	0.86 ± 0.03	0.93 ± 0.02
CIFAR-10 DDPM (40M)	0.68 ± 0.05	0.82 ± 0.04	0.89 ± 0.03
CIFAR-10 DDPM (96M)	0.70 ± 0.04	0.84 ± 0.04	0.91 ± 0.03

Table 9: Final active-mode or active-class count at matched budget $B = 0.10$. GMM uses $K = 10$, $G = 50$, $n = 5000$, and 100 held-out seeds; VaDE and VAE use $G = 50$ and ten held-out seeds; the compact DDPM uses $G = 20$ and five held-out seeds.

Policy	GMM modes	VaDE components	VAE classes	DDPM classes
Geometry-triggered	9.9 ± 0.1	9.3 ± 0.4	9.6 ± 0.3	9.4 ± 0.4
Fixed $\alpha = 0.10$	9.4 ± 0.3	8.2 ± 0.6	8.8 ± 0.5	8.6 ± 0.5
Distribution-triggered	—	7.6 ± 0.7	8.6 ± 0.6	8.3 ± 0.6
Random-triggered	8.7 ± 0.5	6.3 ± 0.8	7.8 ± 0.7	7.5 ± 0.8
Oracle minimum prevalence	10.0 ± 0.0	—	—	—

The VaDE intervention also admits a matched-branch test. We branch from identical checkpoints, use identical subsequent randomness, and consider single geometric alarms separated from other alarms by at least ten generations. One intervention following an alarm reduces component loss over the next ten generations from 62% to 14%. This comparison links geometric warning to a training action and a subsequent reduction in collapse.

For the VAE, final decoder feature rank is 8.0 ± 0.4 under geometric triggering, 7.2 ± 0.5 under fixed $\alpha = 0.10$, 6.9 ± 0.5 under distribution triggering, and 5.8 ± 0.6 under random triggering. The compact-DDPM results in Table 10 show that the coverage advantage is accompanied by better sample quality and preserved representation and curvature ranks.

Table 10: Compact-DDPM intervention endpoints. Feature and GGN ranks are normalized. The final row uses the larger budget $B = 0.25$.

Policy	FID	Entropy	Feature rank	GGN rank
Geometry-triggered	32.8 ± 2.6	2.19 ± 0.05	0.90 ± 0.03	0.85 ± 0.04
Fixed $\alpha = 0.10$	38.7 ± 3.0	2.03 ± 0.06	0.81	0.77
Distribution-triggered	40.5 ± 3.3	1.96 ± 0.07	0.76	0.71
Random-triggered	49.2 ± 4.1	1.78 ± 0.09	0.64	0.60
Fixed $\alpha = 0.25$	29.9 ± 1.0	2.28 ± 0.05	0.96 ± 0.02	0.89 ± 0.03

I.7 Dose scaling and prospective transfer

The GMM guarantee depends on rare-mode prevalence through $\alpha_{\mathrm{GMM}} = \log(KG/\delta)/(n\pi^*_{\min})$. To test its quantitative transfer, we define $\alpha_{\min}$ as the smallest tested dose that preserves every protected mode or class through generation G in at least 90% of held-out runs. Twelve GMM settings vary K, n, G, rare-mode mass, and δ. For neural models, the corresponding predicted scale is

$$s = \frac{\log(K_{\mathrm{eff}} G/\delta)}{n\widehat{\pi}_{\min}},$$

where protected classes or clusters and their prevalences are defined from real data. The neural transfer is an empirical test of the dose scale, not an extension of the labeled-GMM guarantee.

For prospective evaluation, all inputs to the dose rule are measured from the real-data distribution before recursive training. A single multiplicative calibration is selected on GMM/VaDE development settings and transferred without retuning to held-out VAE/DDPM settings. The dose is frozen before held-out evaluation, separating prospective transfer from retrospective dose selection.

Table 11: Quantitative dose-transfer results. All protection rates concern preservation of every protected stratum over the evaluation horizon.

Evaluation	Result
Twelve GMM settings	Log–log slope 0.99, 95% CI $[0.93, 1.05]$, $R^2 = 0.95$
Twelve pooled neural settings	Spearman correlation 0.94; log–log slope 0.93, 95% CI $[0.82, 1.05]$, $R^2 = 0.88$
VaDE / VAE / DDPM slopes	0.95 / 0.91 / 0.88
Protection at $0.5s$ / s / $2s$	47% / 89% / 97% of held-out runs
Factor-of-two agreement	Theory-guided dose within a factor of two of $\alpha_{\min}$ in $11/12$ neural settings
Prospective VAE/DDPM transfer	92% protection at mean $\alpha = 0.15$, versus 93% at fixed $\alpha = 0.25$; 40% fewer real examples

Practical dose selection. Define K_{eff} protected classes or clusters and estimate a conservative lower bound $\widehat{\pi}_{\min}$ on their minimum real-data prevalence. Choose a recursion horizon G, per-generation sample size n, and tolerated failure level δ, then compute

$$\alpha_0 = \min\left\{1, \frac{\log(K_{\text{eff}} G/\delta)}{n\widehat{\pi}_{\min}}\right\}.$$

On development runs, evaluate $0.5\alpha_0$, α_0, and $\min\{1, 2\alpha_0\}$, and select the smallest dose preserving coverage and geometric rank. Freeze the choice for held-out evaluation. If the untruncated expression exceeds one, the sufficient GMM bound does not certify the desired guarantee at the stated n, horizon, and prevalence; clipping makes the dose feasible but does not supply that guarantee.

I.8 Full-model LLC, controls, and intervention

Full-model LLC is estimated in VaDE, the standard VAE, and the 12M-parameter CIFAR-10 DDPM. Synthetic recursion reduces the estimates by approximately 38–41%, whereas matched fixed-real controls remain close to initialization (Table 12). Paired-bootstrap 95% confidence intervals for the recursive generation slopes exclude zero in all three models.

Table 12: Full-model LLC endpoints. The DDPM here is the 12M-parameter CIFAR-10 model.

Model	Initial	Synthetic final	Fixed-real final	$\alpha = 0.25$ final
VaDE	128.4 ± 6.7	80.1 ± 8.9	124.9 ± 5.9	119.6 ± 6.2
VAE	116.2 ± 5.8	69.4 ± 7.6	112.8 ± 6.1	108.9 ± 5.7
DDPM (12M)	214.7 ± 9.8	127.5 ± 11.6	207.9 ± 10.2	198.6 ± 9.4

Table 13: Relative LLC changes and paired differences with paired-bootstrap 95% confidence intervals. Loss prevented is the proportion of the initial-to-synthetic LLC decrease recovered by the $\alpha = 0.25$ intervention.

Model	Synthetic decline	Fixed-real decline	Control–synthetic (95% CI)	Loss prevented
VaDE	37.6%	2.7%	44.8 [38.7, 50.9]	82%
VAE	40.3%	2.9%	43.4 [35.8, 51.2]	84%
DDPM (12M)	40.6%	3.2%	80.4 [68.1, 92.7]	82%

The VAE decoder feature-rank endpoints are reported in Table 14. In VaDE, the componentwise exponent 0.94 with $R^2 = 0.85$ is consistent with the component-mass bound. The value 416 ± 52 is the dimension $65|E_g^{\varepsilon}|$ of the candidate near-null latent-prior subspace at generation 50. Fixed-real training and real-data injection remain close to initial Fisher rank. These results support the singular-complexity interpretation.

I.9 Autoregressive language-model transfer

Recursive OPT-125M training on WikiText-2 tests a different modality and an autoregressive architecture. Synthetic recursion is compared with matched fixed-real training and the same $\alpha = 0.25$

Table 14: VAE decoder feature-rank endpoints.

Diagnostic	Initial	Synthetic final	Fixed-real final	$\alpha = 0.25$ final
VAE decoder feature rank	8.6 ± 0.5	3.7 ± 0.6	8.1 ± 0.4	7.6 ± 0.5

intervention used in the image-model studies. Full-model LLC falls by 38.9% under synthetic recursion, accompanied by higher held-out perplexity and lower coverage of token types in the rarest real-data frequency quintile (Table 15). The fixed-real minus synthetic final LLC difference is 101.7, with paired-bootstrap 95% CI $[86.2, 117.4]$. The LLC-generation slope is negative in all five recursive seeds; paired-bootstrap 95% confidence intervals for both the recursive slope and the recursive-versus-control slope difference exclude zero.

Table 15: OPT-125M on WikiText-2. Hidden-state and GGN ranks are normalized. Rare-token coverage concerns the rarest real-data frequency quintile.

Metric	Initial	Synthetic final	Fixed-real final	$\alpha = 0.25$ final
Full-model LLC	286.3 ± 11.4	174.8 ± 13.7	276.5 ± 10.9	272.1 ± 10.8
Held-out perplexity	21.4 ± 0.6	34.7 ± 1.8	21.9 ± 0.7	22.3 ± 0.8
Rare-token coverage	0.76 ± 0.02	0.41 ± 0.04	0.73 ± 0.02	0.72 ± 0.02
Hidden-state rank	0.91 ± 0.01	0.58 ± 0.03	—	0.88 ± 0.02
Blockwise GGN rank	0.89 ± 0.02	0.61 ± 0.04	—	0.86 ± 0.03

Across checkpoints, LLC has Spearman correlation 0.92 with hidden-state covariance rank, 0.89 with blockwise GGN rank, and 0.90 with rare-token coverage. Thus the LLC decline coincides with selective lexical-tail loss and representation degeneration, rather than only a change in average perplexity. With $\alpha = 0.25$, the final LLC decline is limited to 5.0%, and the representation, curvature, coverage, and perplexity measurements all remain near their initial values.

I.10 Monitoring cost and larger diffusion models

We use localized tempered SGLD with minibatch gradients at sparse checkpoints to estimate LLC. Block-local LLC fixes the remaining parameters. Monitoring overhead and LLC–proxy correlations are shown in Table 16; predictive AUROCs appear in Table 8. The measured monitoring overhead remains below 6% on average through 96M parameters.

Table 16: CIFAR-10 DDPM monitoring cost and block-local LLC changes. Overhead is relative wall-clock cost; LLC–proxy correlations use sparse checkpoints.

Size	Overhead	LLC–proxy	Recursive decline	Control decline
12M	$3.8 \pm 0.4\%$	0.89	—	—
40M	$4.5 \pm 0.6\%$	0.83	34.7%	3.9%
96M	$5.2 \pm 0.7\%$	0.85	31.8%	4.6%

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the paper's main theoretical contributions, including the labeled-mixture RLCT trajectory, the finite-horizon real-data injection guarantee, and the VaDE Fisher-curvature bridge. They also describe the empirical scope accurately by framing the VAE and DDPM results as broader empirical signatures rather than exact theorem-level consequences.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 4 discusses the main limitations: LLC/Hessian/Gauss–Newton diagnostics are more computationally expensive than standard sample-quality metrics. The discussion also identifies multimodal models and broader settings as future work.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Section 2 states the assumptions for the labeled Gaussian-mixture recursion, the separated-survivor RLCT regime, the finite-horizon real-data injection result, and the VaDE Fisher-curvature bridge. Proof sketches are given in the main text, with full proofs and derivations provided in the appendix for the Aoyagi–Nagata monotonicity property, the RLCT trajectory theorem, the real-data injection bound, and the neural Fisher/decoder curvature identities.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 3 describes the recursive training protocol, datasets, model classes, seeds, sample sizes, horizons, intervention levels, and evaluation metrics used for the main experiments. Additional architecture, optimizer, estimator, bootstrap, and compute details are provided in the appendix, including the LLC/SGLD estimator settings and robustness checks.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.

(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: We provide an anonymized zip file containing the code, configuration files, figure-generation scripts, and instructions needed to reproduce the main experiments. The supplemental material will describe data access and preparation for the public datasets used, including MNIST and FashionMNIST, as well as the commands needed to run the experimental pipeline.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 3 specifies the experimental tiers, recursive training protocol, datasets, sample sizes, horizons, seeds, intervention levels, and evaluation metrics. Full architecture, optimizer, hyperparameter, estimator, and compute details are provided in the appendix and supplemental material.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Section 3 states that curves report mean $\pm$ one standard deviation over 10 seeds, with 5 seeds for the DDPM tier. The figure captions also report 1σ bands or error bars where applicable, and additional bootstrap/error-bar details are provided in the appendix.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The appendix reports the compute resources used for the Gaussian-mixture, VaDE, VAE, and DDPM experiments, including the relevant CPU/GPU setting, memory requirements, and approximate runtime. The supplemental material also provides the commands and configuration files needed to reproduce the reported runs.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

Justification: The work uses public benchmark datasets and synthetic/generated data for methodological evaluation, and does not involve human-subjects research, private user data, surveillance, or deployment in a high-stakes setting. The paper is anonymized for double-blind review and discusses limitations and scope.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [No]

Justification: The paper is primarily foundational and methodological, focusing on diagnostics for recursive generative training rather than a deployed system or application domain. Potential positive impacts include improving reliability and monitoring of self-consuming generative pipelines, while potential negative impacts are indirect and would depend on downstream use of generative models; these impacts are not discussed in detail in the current manuscript.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release high-risk pretrained models, scraped datasets, or deployed generative systems. The released materials are intended to support reproducibility of small-scale benchmark experiments on public datasets such as MNIST and FashionMNIST, so no special controlled-release safeguards are required.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The paper uses public benchmark datasets, including MNIST and Fashion-MNIST, and standard open-source scientific computing libraries; these assets are credited in the paper or supplemental material. The supplemental material includes dataset access information, relevant license/terms-of-use information where available, and the released code package will include license and attribution information.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The paper will release an anonymized code package through Google Drive containing experiment scripts, configuration files, figure-generation code, and reproduction instructions. The released materials will include documentation describing the training settings, datasets, licenses, limitations, and commands needed to reproduce the main results.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.

- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing, human-subjects experiments, participant instructions, or participant compensation. All experiments use synthetic data or public benchmark datasets such as MNIST and FashionMNIST.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing, human-subjects research, participant data collection, or interventions involving study participants. All experiments use synthetic data or public benchmark datasets such as MNIST and FashionMNIST.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: The core method development, theoretical results, and experiments in this paper do not use LLMs as an important, original, or non-standard component. Any LLM use was limited to writing, editing, or formatting assistance and did not affect the scientific methodology, experimental design, or originality of the work.

    Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.